DSRG
DIGITAL SOCIETY RESEARCH GRANT

DSRGX
DIGITAL SOCIETY RESEARCH GRANT FOR INDUSTRY

DIGITAL SOCIETY RESEARCH GRANT (DSRG) CYCLE 2 & DIGITAL SOCIETY RESEARCH GRANT FOR INDUSTRY (DSRGX) 2025

APPLICATION GUIDELINE

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TABLE OF CONTENTS

No.	Title	Page
SECTION 1: ABOUT THE DIGITAL SOCIETY RESEARCH GRANT (DSRG) AND DIGITAL SOCIETY RESEARCH GRANT FOR INDUSTRY (DSRG*X*)		**2**
1.1	Introduction	2
1.2	Objective	3
1.3	Frequency	4
1.4	Research Focus Areas	4
1.5	Duration of Project	11
1.6	Grant Amount	12
1.7	Matching Grant/Additional Fund	12
SECTION 2: APPLICATION PROCESS AND PROCEDURES		**12**
2.1	Eligibility Criteria	12
2.2	Research Team	12
2.3	Research Proposal	13
2.4	Expenditure Details	13
2.5	Proposal Submission	14
2.6	Evaluation of Proposals	15
2.7	Award/Rejection of Proposals	16
2.8	Submission, Evaluation and Award Process Timeframe	17
SECTION 3: PROJECT IMPLEMENTATION AND MONITORING		**19**
3.1	Disbursement of Funds	19
3.2	Submission of Reports, Slides and Project Manuscript	19
3.3	Dissemination of Findings	21
3.4	Project Closure Notification	21
3.5	Project Implementation and Monitoring Process Flow	21
SECTION 4: INTELLECTUAL PROPERTY		**23**
4.1	Intellectual Property (IP)	23
4.2	Publishing Rights	23
APPENDIX 1: DSRG CYCLE 2 AND DSRG*X* 2025 RESEARCH TITLES		**24**

SECTION 1: ABOUT THE DIGITAL SOCIETY RESEARCH GRANT (DSRG) AND THE DIGITAL SOCIETY RESEARCH GRANT FOR INDUSTRY (DSRG*X*)

1.1 Introduction

1.1.1. The Malaysian Communications and Multimedia Commission (MCMC) Digital Society Research Grant (DSRG) was conceived to contribute toward enhancing necessary information resources and in line with changing community expectations as we navigate the transition towards a sustainable digital civil society.

1.1.2. Beyond the provisioning of infrastructure and communications services, it is imperative that users possess the knowledge, skills, and attitudes to harness the potential of digital media and communications effectively in meeting these aspirations. At its most basic element, users ought to be aware of the many opportunities that the digital age affords its users, from access to information, education, electronic services of all sorts, and the potential for socio-economic opportunities.

1.1.3. Accompanying these connectivity benefits is also the potential for risk, whether from fraudulent or criminal activities, online harm, cyberbullying, or predation. Thus, users must ensure that they build their resilience and understanding of these harms. Accordingly, digital media literacy has increasingly become a key competency for citizen and user participation across the digital economy and for society in the twenty-first century.

1.1.4. Moving up the value chain, another important aspect of knowledge, skills, and attitude is the new work competencies that need to be acquired. These may take the form of general work skills such as the sophistication to work in digital environments or alternatively to acquiring specialist technical knowledge for new work opportunities brought about by new services or technologies, which include 5G, AI, 5IR and the ongoing digitalisation of all industry verticals.

1.1.5. Thus, society's heightened dependence on digital technologies has made it necessary for us to continuously address awareness and literacy and acquire the new competencies necessary for the future of work. More than ever, coherent and intelligent insights are required to address inequitable opportunity, access, knowledge, and skill issues. The efforts must be directed at ensuring the readiness and resilience of communities as Malaysia strives to keep

abreast of global trends and ensure our continued competitiveness and resilience.

1.1.6. In addressing these concerns, we are guided by MCMC's National Policy Objectives to promote a civil society where information-based services will provide the basis of continuing enhancements to the quality of work and life.

1.1.7. In addressing the research gaps, research outcomes aim to support the strategies and initiatives under the various ongoing National Plans. These include the *Pelan Jalinan Digital Negara* (JENDELA), the Malaysia Digital Economy Blueprint (MyDIGITAL), the National 4th Industrial Revolution (4IR) Policy, the 12th Malaysia Plan (RMK-12), and the Malaysian Budget of 2025.

1.1.8. The Digital Society Research Grant for Industry (DSRG*X*) is an extension of the Digital Society Research Grant (DSRG), specifically designed to address industry-specific gaps and foster collaboration among industry players and stakeholders.

1.1.9. Launched to complement the DSRG, the DSRG*X* aims to generate evidence-based research that is directly applicable to the Communications and Multimedia (C&M) industry, ensuring that the findings benefit all industry players and stakeholders.

1.1.10. By focusing on industry-specific challenges and opportunities, DSRGX supports the development of policies, programmes, and interventions that promote digital inclusion, enhance industry knowledge, and drive collective progress.

1.2 Objective

1.2.1. This grant aims to grow the evidence base necessary for the nation to optimise the advancements made in communications infrastructure and service deployment. This base will assist the development of policy, programmes, and interventions to promote the inclusion and participation of all population segments as the nation transitions towards being a fully digitally connected and informed society.

1.3 Frequency

1.3.1. For 2025, the first cycle of the DSRG was issued in March, followed by a second cycle in August. The only cycle for DSRGX started in

August, coinciding with the issuance of DSRG Cycle 2. Both initiatives are planned to be held annually.

1.4 Research Focus Areas

1.4.1. The research proposals are guided by the following four (4) Focus Areas:

i. Digital Citizenship & Cyberwellness (DCC): Aims to elicit research clarifying regulatory and developmental gap areas influencing participation, positive uptake and wellness in an increasingly digitally dependent world;

ii. Digital Inclusion (DI): Seeks to clarify regulatory and developmental gaps in areas related to factors that impede equality of access and challenge the paradigm of ensuring that no one is left behind or is deprived of digital connectivity and its benefits;

iii. Future of Work (FW): Aims to clarify gap areas related to work transformations and the evolving skills needed due to digitalisation, technological advances and changing workforce expectations, and develop strategies to equip individuals and organisations for success; and

iv. Sustainability & Environment, Social, and Governance (SESG): Aims to clarify gap areas impacting the environmental footprint of technology, social and ethical factors, sustainable business practices, and initiatives in the C&M industry and supporting national Sustainable Development Goals.

Table 1 – Framework for DSRG Cycle 2 and DSRG*X* 2025 Focus Areas

No.	Research Focus Areas	Sub-focus areas
1	**Digital Society and Cyberwellness (DCC)**	1. Competencies and literacies 2. Cybersecurity risks 3. User rights 4. Consumer protection 5. Awareness and self-regulation
2	**Digital Inclusion (DI)**	1. Empowering excluded groups 2. Addressing the digital divide 3. Access to health and assisted living services
3	**Future of Work (FW)**	1. Job market trends 2. Skills and reskilling 3. Flexible work models 4. Leadership 5. AI literacy 6. Organisational change
4	**Sustainability & Environment, Social, and Governance (SESG)**	1. Environmental Impact of Technology ○ Energy efficiency and carbon reduction ○ E-waste management ○ Green supply chain and initiatives ○ Sustainable infrastructure and operations 2. Ethical and Responsible Technology Use ○ Responsible use of AI and other emerging technologies ○ Data ethics and algorithmic transparency ○ Risk management and regulatory compliance 3. Sustainable Business and Governance Practices ○ ESG integration in the C&M industry ○ Corporate governance and stakeholder accountability ○ Public trust and institutional integrity

1.4.2. For DSRG Cycle 2 and DSRG*X* 2025, interested researchers are invited to submit project proposals on one of the 21 research titles below:

List of Guided DSRG and DSRG*X* Research within the Digital Citizenship and Cyberwellness (DCC), Digital Inclusion (DI), Future of Work (FW), and Sustainability & Environment, Social, and Governance (SESG) Categories

No.	Code	Research Title	Research Category and Gap/Developmental Area
1	DCC-1	**The Content Code and the General Consumer Code: Assessing Their Awareness, Relevance, Usability, and Perceived Benefits Among the Malaysian Public in the Communications and Multimedia Industry**	Falling under the Guided Research Category of Digital Citizenship and Cyberwellness, this study aims to generate evidence that will aid in the validation and improvement of awareness and self-regulation of the identified industry codes on the Malaysian public.
2	DCC-2	**Evaluating *Kempen Internet Selamat*: Developing a Standardised Framework for Monitoring, Enhancing, and Measuring Online Safety Impact**	Falling under the Guided Research Category of Digital Citizenship and Cyberwellness, this study aims to generate evidence that will aid in the validation and improvement of awareness and self-regulation of *Kempen Internet Selamat* (KIS).
3	DCC-3	**Strengthening Public Adoption of Consumer Redress Mechanisms in the Communications, Multimedia, and Postal Services Industry: A Case Study in Negeri Sembilan**	Falling under the Guided Research Category of Digital Citizenship and Cyberwellness, this study aims to generate evidence that will aid in the awareness and adoption consumer redress mechanisms serving users of communications, multimedia and postal services industry.
4	DCC-4	**Cashless Payment Adoption in Negeri Sembilan: A Case Study of the General Public and Government Sector under the MCMC Cashless Society Scaling Up Programme**	Falling under the Guided Research Category of Digital Citizenship and Cyberwellness, this study aims to generate evidence on the MCMC Cashless Society Scaling Up Programme and examines general public and government sector readiness, challenges, and strategic opportunities to support inclusive digital transformation.

No.	Code	Research Title	Research Category and Gap/Developmental Area
5	DCC-5	**Understanding Digital Content Access Norms: Behavioural Archetypes and Ethical Perspectives Among Young Citizens**	Falling under the Guided Research Category of Digital Citizenship and Cyberwellness, this study aims to generate evidence related to digital content access norms amongst young citizens.
6	DI-1	**Investigating the Influence of Industrialised Building System (IBS) Precast Technology on Indoor Cellular Network Performance in Selangor Residential Developments**	Falling under the Guided Research Category of Digital Inclusion (DI), this study aims to generate evidence to contribute to the development of policies and guidelines to improve the quality of in-building cellular connectivity.
7	DI–2	**Assessing Langkawi's Digital Infrastructure for Inclusive Health, Assisted Living, and Wellness Tourism**	Falling under the Guided Research Category of Digital Inclusion (DI), this study aims to generate evidence to contribute to the development of programme evaluation, assessment and impact of efforts to improve access to health, assisted living, wellbeing and health tourism services.
8	DI-3	**Digital Skills Training at NADI: Assessing AI Tutors and Human Educators for Future Workforce Readiness**	Falling under the Guided Research Category of Digital Inclusion (DI), this study aims to generate evidence to address the digital divide and to ensure access to quality digital education and skills.
9	DI-4	**An Assessment of the Functionality and Connectivity of Master Antenna Television (MATV) Systems in Affordable High-Rise Residential Buildings in the Central Region**	Falling under the Guided Research Category of Digital Inclusion (DI), this study aims to generate evidence to address the digital divide by evaluating connectivity and its impact on access to broadcast information.

No.	Code	Research Title	Research Category and Gap/Developmental Area
10	DI-5	**Aligning Connectivity and Development: Evaluating JENDELA's Impact in Rural Sabah Communities with Infrastructure Gaps**	Falling under the Guided Research Category of Digital Inclusion, this research aims to generate evidence on how JENDELA-enabled connectivity in rural Sabah influences socio-economic impact amid ongoing infrastructure gaps.
11	DI-6	**Positioning 5G as a Public Utility in Sabah: Strategies for Inclusive and Impactful Connectivity**	Falling under the Guided Research Category of Digital Inclusion, this research aims to generate evidence to support equitable connectivity by identifying strategies and measures for 5G adoption in Sabah. The study focuses on addressing the digital divide, particularly among underserved communities, in ways that enable meaningful participation in digitalisation and society.
12	DI-7	**Enabling FTTH Adoption in Brownfield Zones: A Sabah Case Study on Infrastructure Diversity and Consumer Choice**	Falling under the Guided Research Category of Digital Inclusion, this research aims to generate evidence to support digital inclusion by identifying strategies and measures for FTTH adoption in Sabah. The study focuses on factors for ensuring equitable connectivity and addressing the digital divide, and to provide insights to ensure the sustainability on non-incumbent infrastructure providers.
13	FW-1	**Building a Sustainable Future of Work: Assessing Flexible Working Hours in the Sabah Telecommunications Sector**	Falling under the Guided Research Category of Future of Work, this research aims to generate evidence that will aid in the assessment of the adoption of Flexible Working Arrangements (FWA) within the telecommunications sector.

No.	Code	Research Title	Research Category and Gap/Developmental Area
14	SESG-1	**Catalysing ESG Integration in the Communications and Multimedia Industry: MCMC's Role in Policy Development and Best Practice Facilitation**	Falling under the Guided Research Category of Sustainability & Environment, Social, and Governance (SESG), this study aims to generate evidence that will aid in the development of MCMC policies and facilitation of ESG adoption amongst C&M industry players.
15	SESG-2	**Transforming the Malaysian Communications and Multimedia Designated Industry Fora: A Governance and Operational Review**	Falling under the Guided Research Category of Sustainability & Environment, Social, and Governance (SESG), this study aims to generate evidence that will aid in the development of MCMC policies supporting and facilitating the adoption of SESG adoption within the C&M designated industry fora.
16	SESG-3	**Enhancing the Financial Sustainability of Designated Industry Fora in Malaysia: Exploring Alternative Mechanisms for Long-Term Viability**	Falling under the Guided Research Category of Sustainability & Environment, Social, and Governance (SESG), this study aims to generate evidence that will aid in the development of relevant policies supporting and facilitating corporate governance and stakeholder accountability within the designated industry fora.
17	SESG-4	**A Study on Public Utility Perception and Legal Safeguards for Fibre Infrastructure in Sabah: Addressing Vandalism, Theft, and Community Awareness**	Falling under the Guided Research Category of Sustainability & Environment, Social, and Governance (SESG), this research contributes to the Sustainability and ESG agenda by examining governance mechanisms, community engagement, and legal safeguards that support the long-term resilience and public

No.	Code	Research Title	Research Category and Gap/Developmental Area
			trust in Sabah's telecommunications infrastructure.
18	SESG-5	**SW110 Waste Management in Sabah: Bridging Infrastructure Gaps and Evaluating Localised Treatment Feasibility**	Falling under the Guided Research Category of Sustainability & Environment, Social, and Governance (SESG), this research contributes to the Sustainability and ESG agenda by critically examining governance mechanisms, policy frameworks, and legal provisions surrounding e-waste management in Sabah, highlighting gaps between regulation and real-world enforcement.
19	SESG-6	**Evaluating the Relevance of Regulatory Definitions in the General Consumer Code (GCC) Amidst Malaysia's Evolving Communications and Multimedia Landscape**	Falling under the Guided Research Category of Sustainability & Environment, Social, and Governance (SESG), this study aims to determine the relevance of the General Consumer Code of Practice for the Communications and Multimedia Industry Malaysia (GCC) developed by the CFM in ensuring sustainable business and consumer practices through sound corporate governance and stakeholder accountability.
20	SESG-7	**Strengthening Business Stability in Malaysian Broadcasting through Sustainable Local Content Creation: A Strategic Framework for Producers and Regional Broadcasters**	Falling under the Guided Research Category of Sustainability & Environment, Social, and Governance (SESG), this study aims to understand the factors contributing to and impeding the sustainability of local Malaysian content creation towards ensuring the sustainability of Malaysia

No.	Code	Research Title	Research Category and Gap/Developmental Area
			broadcasters and local content creators.
21	SESG-8	**Closing the Data Gaps: Developing Locally Representative Emission Factors for Malaysia's Transportation Sector to Strengthen Climate Policy and Reporting**	Falling under the Guided Research Category of Sustainability & Environment, Social, and Governance (SESG), this study contributes to the SESG agenda by examining environmental emission factors and measurement methodologies relevant to industries heavily dependent on transportation (e.g., logistics, delivery services), with a focus on the postal and courier sector's transition toward low-carbon fleet operations.

1.4.3. The above research titles are categorised as **Guided Research**, where the predetermined Research Objectives (ROs) are to be achieved with researchers expected to propose a research design incorporating suitable theoretical or conceptual frameworks, development of research questions, research instruments and methodology.

1.4.4. Please refer to **Appendix 1** for further information on the gap area, targeted research subjects, research problem and context, and desired research aims and objectives.

1.5 Duration of Project

1.5.1. The term of a project is up to nine (9) months, comprising six (6) months of research activities until the submission of the research report at the end of the sixth month and three (3) months of administrative closure.

1.5.2. The project shall be completed according to the duration and deadlines stipulated in the LOA. Applicants shall indicate the project duration in the proposal, including each phase of work.

1.6 Grant Amount

1.6.1. The grant amount shall depend on the type and scope of the research project and subject to the guidelines herein and may be of a sum of up to Ringgit Malaysia ten thousand (RM10,000).

1.7 Matching Grant/Additional Fund

1.7.1 Researchers can source matching grants/additional funds from their universities or other funding bodies for the expenses not covered by DSRG Cycle 2 or DSRG*X* 2025, or as additional funding for their projects.

SECTION 2: APPLICATION PROCESS AND PROCEDURES

2.1 Eligibility Criteria

2.1.1. The grant is open to full-time academic faculty members in schools of communications, social sciences, humanities, or related fields of private and public Higher Educational Institutions (HEI). Each proposal must have a Lead Researcher, subject to the general terms and conditions for granting.

2.1.2. The following rules apply to the applicant:

i. Lead Researcher must hold a doctoral degree; (PhD or equivalent);
ii. Lead Researcher must have an appointment with a local HEI for (at least) the duration of the proposed research project;
iii. The salary of the researcher(s) cannot be financed from this grant;
iv. The researcher(s) may request the grant on her/his behalf and on behalf of any possible project consortium; and
v. The researcher(s) is responsible for research and financial matters.

2.1.3. Researcher(s) can only submit one (1) proposal as the Lead Researcher within this call, and each researcher can act no more than twice as an applicant (as Lead Researcher or Co-Researcher).

2.2 Research Team

2.1.1 The research team **must comprise at least two (2) researchers (a Lead Researcher and a Co-Researcher)**.

2.1.2 Researcher(s) in professions other than academia are allowed to join the research team as co-researchers to complement the expertise and with the expectation that the research product will contribute to the broader body of knowledge on the topic specified.

2.1.3 The MCMC officers (including the research sponsors) cannot be included as co-researchers.

2.1.4 Research assistants paid from the awarded DSRG Cycle 2, 2025 or DSRG*X* grant cannot be included as members of the research team. Additionally, research team members cannot be paid for any professional services for the research work from the awarded amount.

2.3 Research Proposal

2.1.5 The DSRG Cycle 2 and DSRG*X* 2025 proposal submission form is available for download on the MCMC website.

2.1.6 The research proposal must also consider and include a contingency plan for disruptions to mitigate such risks. Any requests for extension of project deadlines are discouraged, and all reasonable attempts must be made to preserve the timely completion of deliverables.

2.4 Expenditure Details

2.4.1. **Remuneration and Allowances**
The wages and allowance are only for temporary and contract personnel directly engaged in the project. The period of employment and hourly/monthly rate for the research assistant(s) must be clearly stated and justified.

2.4.2. **Travel and Transportation**
Only travel expenses (domestic) directly related to the project are claimable.

2.4.3. **Rental**
Only rental expenses for building space, equipment, transportation and any other item(s) directly related to the project are claimable.

2.4.4. **Research materials and supplies**
Only extends to expenses for research materials and supplies directly related to the project, such as books, magazines, computer

software, photocopying, printing, binding, filming, consumables (stationeries, etc.), charges from postage, telephone, fax and other expenses necessary to complete the project. The purchase of assets and electronic gadgets, such as computers, tablets, phones, printers, etc., is not claimable.

2.4.5. **University Management Fees**
Payment made to Lead Researcher's HEI to conduct this research (if required).

2.4.6. **Special Services**
Translation, license for Grammarly, incentives for data collection, data gathering, and processing costs are claimable.

2.4.7. **Proofreading and editorial services**
Payment made specifically for qualified proofreading services for the research report and project manuscript. This expenditure is not to be used as payment to research members or assistants.

2.4.8. **Conference**
Specifically, to defray conference costs for the Lead Researcher to acquire related knowledge on research or disseminate research findings.

2.4.9. **Publication**
Specifically, to defray costs for publications in relevant academic, indexed and/or peer-reviewed journals.

2.5 Proposal Submission

2.5.1. The proposal(s) verified by the respective HEI Research Management Centres (RMCs) may be submitted in English or Malay and shall be presented clearly and submitted together with the following:

i. Curriculum vitae of the Lead Researcher and team member(s) involved;
ii. Certified true copies of the highest academic certificates;
iii. A copy of the latest published research paper or article by the Lead Researcher; and
iv. Other relevant materials to support the proposal.

2.5.2. If the HEI does not have an RMC, the department or unit responsible for functions similar to an RMC may verify the research grant-related documents.

2.5.3. The electronic copy of the proposal and other documents should be emailed to the Secretariat with **'DSRG/DSRGX 2025 SUBMISSION'** in the subject line and addressed to dsrg@mcmc.gov.my **no later than 5:00 pm, Monday, 22 September 2025**.

2.5.4. An acknowledgement receipt will be sent once the Secretariat has received the proposal. Those who have submitted the proposals and have NOT received an email confirmation within seven (7) working days should contact the Secretariat.

2.5.5. All applicants are advised to adhere to the stipulated requirements. Submissions that do not follow the criteria will risk being disqualified from consideration. **Submissions received after the deadline will not be considered.**

2.6 Evaluation of Proposals

2.6.1. The proposals will be evaluated by the DSRG Cycle 2 and DSRG*X* 2025 Technical Panel based on open competition and merit, and taking into consideration the following criteria:

i. **Quality**: Rationale and justification are presented coherently and logically within the research focus and key growth areas. Ethical considerations have also been identified and addressed;

ii. **Impact of research**: The research problem analysis identified an opportunity to contribute to implementing or evolving one or more communications and multimedia industry's policies or initiatives. The proposed study is also potentially significant for offering new insights into the subject area and other relevant sectors;

iii. **Alignment to internal requirement**: The need and relevancy of the research in contributing towards to organisational or departmental specific works and potentially provides valuable and relevant data for the knowledge base;

iv. **Suitability of applicant**: The degree to which the researchers have the experience, expertise, skills and

knowledge in the proposed area of research and with the proposed methodology to accomplish the stated aims of the project; and

v. **Feasibility**: The appropriateness of the proposed activities, methods, planned activities, and resources to accomplish the project within the stated timeframe. The proposal also identifies the project's challenges and measures to overcome those challenges.

2.6.2. The Lead Researcher may be invited to present their proposal to the Technical Panel as part of the evaluation process.

2.7 Award/Rejection of Proposals

2.7.1. The various factors contributing to the poor suitability of submitted proposals include the following aspects:

i. Researcher(s) do not understand the communications and multimedia industry and MCMC's role and functions, thereby submitting proposals outside of MCMC's regulative scope or too remote in impacting key regulatory partners or stakeholders;
ii. The proposed research is based on the study of research questions with existing high research work and publications and does not provide new insights, value or new knowledge;
iii. Research scope may not be feasible given the grant amount and limited duration allowed for under the DSRG;
iv. Researchers' expertise does not match the research field of the proposal and/or lacks past research experience in the proposed area of study;
v. The literature review and theoretical and/or conceptual frameworks underpinning a proposed study were not included; and
vi. Submissions were of non-research proposals, such as those focusing on prototype, applications or software development.

2.7.2. The Technical Panel reserves the right to consider any other factors it may deem relevant in the evaluation process and to reject proposals that do not meet the submission and evaluation criteria.

2.7.3. Successful applicants will be informed via email. The Technical Panel may suggest changes to the proposals, including cost/funding, scope, and research timelines. The successful applicant must sign the Letter of Award (LOA) and return it to the

Secretariat to indicate acceptance of the grant and its terms and conditions.

2.8 Submission, Evaluation and Award Process Timeframe

2.8.1. An overview of the DSRG Cycle 2 and DSRG*X* 2025, timeframe is tabulated below:

Table 3 – DSRG Cycle 2 and DSRGX 2025 Timeframe

No.	Process	Tentative Date*
i.	Issuance of Call for Proposal	22 August 2025
ii.	Deadline for submissions	5:00 pm, 22 September 2025
iii.	Information session with researchers	The fourth week of August 2025
iv.	Evaluation process	The fourth week of September 2025 to the third week of October 2025
v.	Notification period	Fourth/Fifth week of October 2025
vi.	Execution of LOA	
vii.	Project commencement	First/Second week of November 2025

**Timeline may be affected by public holidays*

2.8.2. Figure 1 below shows the flowchart for the submission, evaluation, and award process of DSRG Cycle 2 and DSRG*X* 2025:

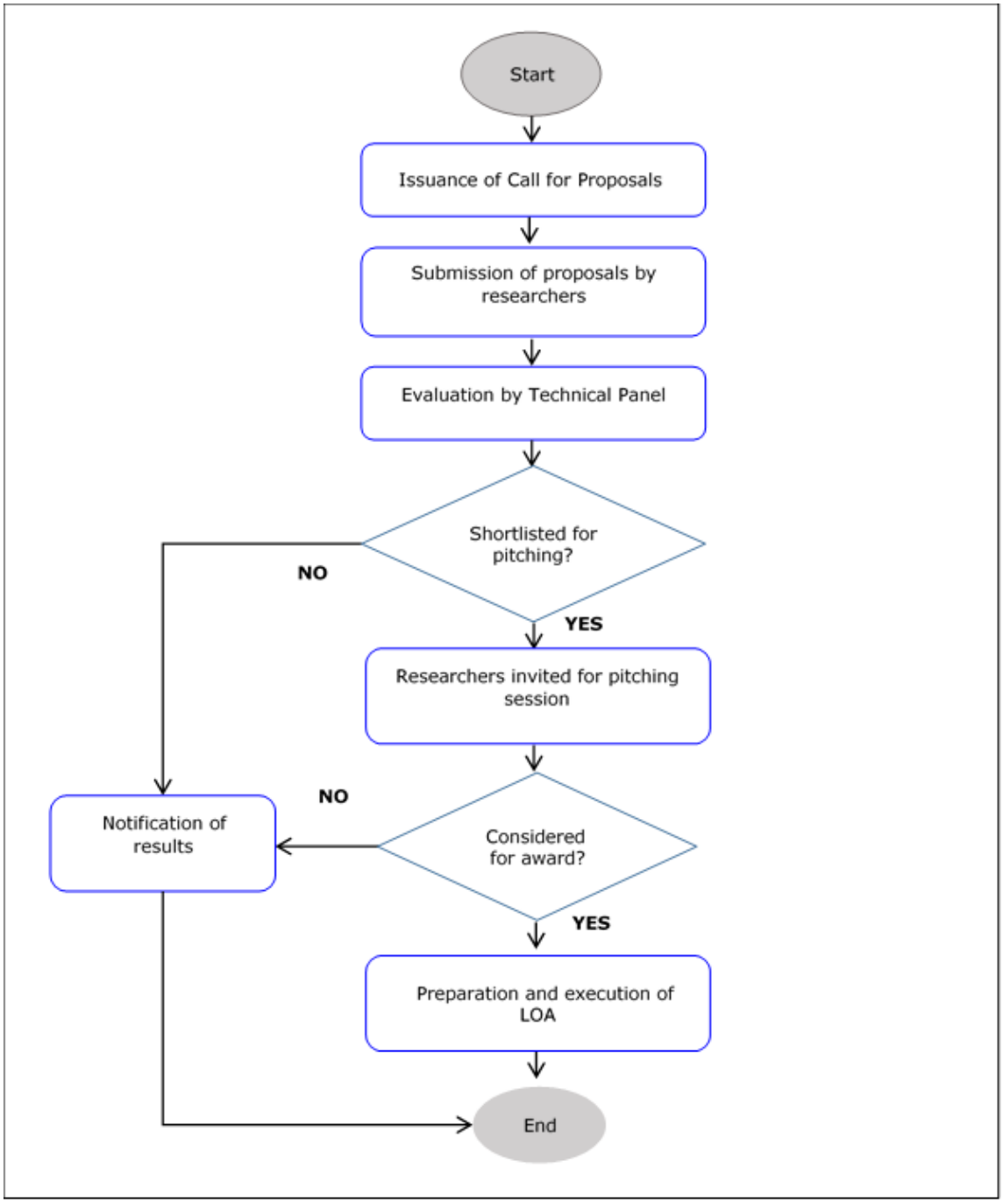


Figure 1 – Process Flow Chart for Submission, Evaluation, and Award of DSRG Cycle 2 and DSRG*X* 2025

SECTION 3: PROJECT IMPLEMENTATION AND MONITORING

3.1 Disbursement of Funds

3.1.1. The grant will be disbursed according to the following schedule:

Table 4 – Grant Disbursement Schedule

No.	Disbursement Phase	Description	Quantum (%)
i.	First disbursement	Upon submission of the signed Letter of Award to the Commission[1]	50
ii.	Second disbursement	Upon submission of the verified Interim Report, subject to the satisfaction of the Commission (*disbursement subject to the Researcher's request*)	40
iii.	Final disbursement	Upon submission of the verified Research Report and no later than one (1) month after the completion of research activities, subject to the satisfaction of the Commission (*disbursement subject to the Researcher's request*)	10

3.2 Submission of Reports, Slides and Project Manuscript

3.2.1. Interim Report (IR)

i. The Lead Researcher is responsible for successfully implementing the project according to agreed timelines and for the timely submission of the IR. It is required for the IR to be submitted at the end of the 3rd month from the project commencement date and upon achieving 50 per cent of the project completion target;

ii. The IR must be verified by the Lead Researcher HEI and submitted together with the financial status update as per the templates provided by the Secretariat; and

iii. The reports will be evaluated against the deliverables to determine whether the project is on track and whether the conditions for disbursement are met.

3.2.2. Research Report (RR)

[1] The signed Letter of Award will then be submitted to the Corporate Advisory Department, Malaysian Communications and Multimedia Commission for the stamping process.

i. The RR must be verified by the Lead Researcher HEI RMC and submitted by the end of the 6th month from the project commencement date and upon achieving 100 per cent of the project completion target.

ii. The RR shall include (but is not limited to) the following and per the template provided by the Secretariat:

- Abstract;
- Introduction;
- Research Objectives (ROs);
- Literature Review;
- Methodology;
- Findings;
- Direct outputs of the research;
- Achievements based on the original ROs;
- Implications and recommendations for regulatory and policy considerations; and
- Recommendations for future research.

iii. The updated and finalised RR based on the feedback and comments provided must be submitted within the 8th month of the project commencement date.

3.2.3. Presentation Slides

i. The updated and finalised presentation slides based on the final RR must be submitted within the 8th month from the project commencement date.

3.2.4. Project Manuscript[2]

i. The project manuscript will be published in MCMC's research publication known as Media Matters;

ii. The project manuscript is required to be submitted as per the templates provided by the Secretariat after the RR is approved;

iii. Researchers who have conducted the research and submitted the reports in Malay shall translate and submit the project manuscript in English.

[2] The project manuscript is an abridged version of the Final Research Report which will be published as part of MCMC's Media Matters.

3.2.5. Financial Report (FR)

i. The FR is to be submitted by the end of the 9th month from the project commencement date, as per the templates provided by the Secretariat and supported with a verified financial statement from the HEI.

3.3 Dissemination of Findings

3.3.1. The researcher(s) will be invited to present their findings at MCMC meetings/seminars/symposiums and may be invited to participate in media engagement activities arranged by MCMC as spokespersons for the research project.

3.3.2. The MCMC may elect to publish and distribute all or portions of the research report and/or project manuscript without restriction.

3.3.3. The researcher(s) may publish the "spin-off" version of the research findings in peer-reviewed journals. The researcher(s) shall ensure the following statement of acknowledgement is clearly stated in the publication report:

*"**This Project is funded by the Malaysian Communications and Multimedia Commission through the Digital Society Research Grant or Digital Society Research Grant for Industry** "*

3.3.4. The researcher shall submit a written notification to the Secretariat before the publication.

3.4 Project Closure Notification

3.4.1. An acknowledgement receipt of project closure will be sent to the Lead Researcher once the requirement for proper project closure and conditions, such as satisfactory submissions of reports and financial statements, are met.

3.5 Project Implementation and Monitoring Process Flow

3.5.1. An overview of the project implementation and monitoring process is provided in Figure 2 below:

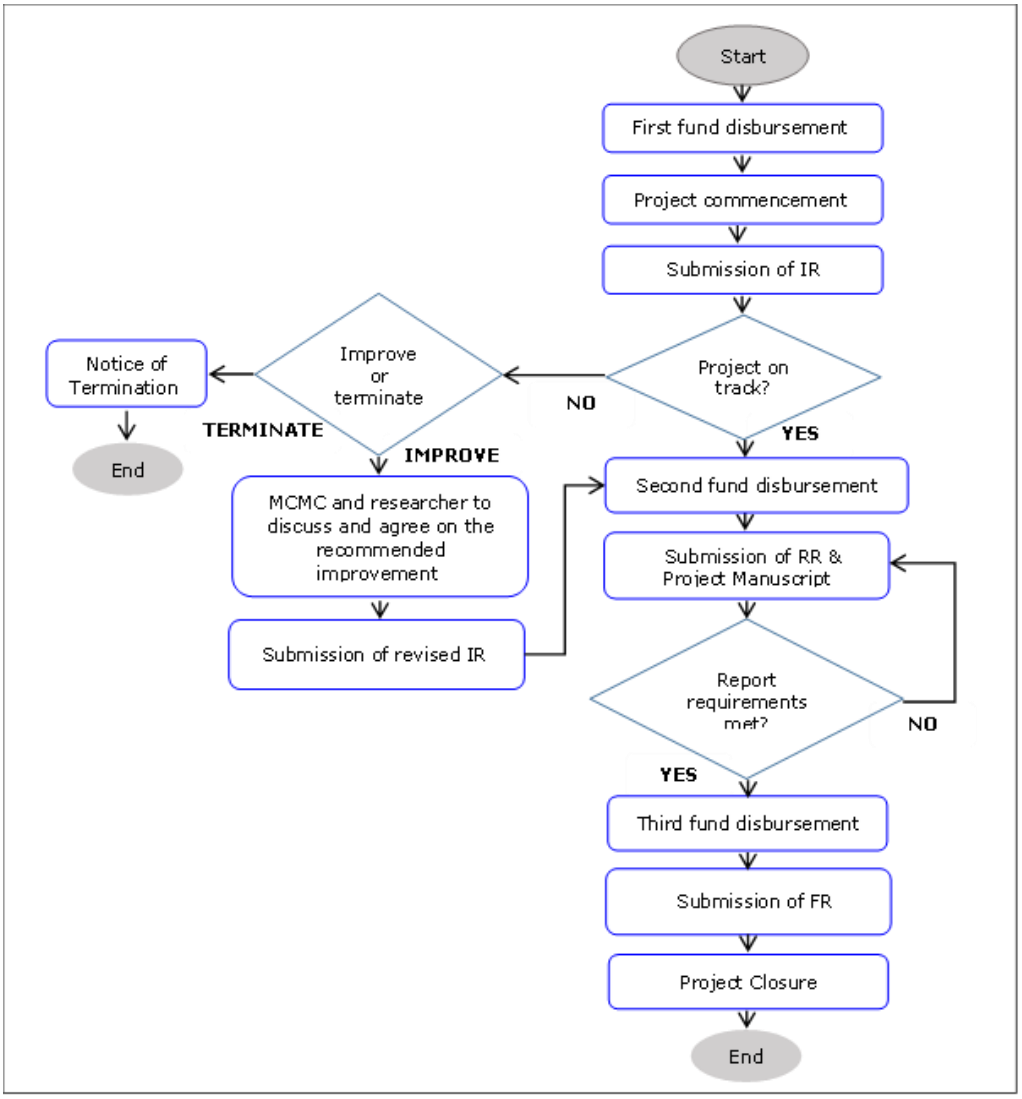


Figure 2 - Process Flow Chart for Project Implementation and Monitoring

SECTION 4: INTELLECTUAL PROPERTY

4.1 Intellectual Property (IP)

4.1.1. Ownership and management of IP, royalties, and any other fees received by the institution resulting from the findings or outputs of the research, such as licensing of the IP or any other forms of commercialisation, shall be governed per the agreed terms and conditions outlined in the LOA.

4.2 Publishing Rights

4.2.1. The MCMC is entitled to publish the research reports in any form deemed fit for education or knowledge transfer. Notwithstanding, the Lead Researcher is required to contribute through publishing and presenting research findings in local or international events/media, subject to the prior approval of the MCMC. Copies of all publications and proof of participation, i.e. presentation slides, write-up, event agenda, etc. are to be submitted to the Secretariat.

4.2.2. The Lead Researcher shall denote and acknowledge the source of research funding and support for the project and the contribution of the various entities.

-End of the Document-

APPENDIX 1:
DSRG CYCLE 2 and DSRGX 2025
RESEARCH TITLES

Framework for DSRG Cycle 2 and DSRGX 2025

No.	Research Focus Areas	Sub-focus areas
1	**Digital Society and Cyberwellness (DCC)**	1. Competencies and literacies 2. Cybersecurity risks 3. User rights 4. Consumer protection 5. Awareness and self-regulation
2	**Digital Inclusion (DI)**	1. Empowering excluded groups 2. Addressing the digital divide 3. Access to health and assisted living services
3	**Future of Work (FW)**	1. Job market trends 2. Skills and reskilling 3. Flexible work models 4. Leadership 5. AI literacy 6. Organisational change
4	**Sustainability & Environment, Social, and Governance (SESG)**	1. Environmental Impact of Technology o Energy efficiency and carbon reduction o E-waste management o Green supply chain and initiatives o Sustainable infrastructure and operations 2. Ethical and Responsible Technology Use o Responsible use of AI and other emerging technologies o Data ethics and algorithmic transparency o Risk management and regulatory compliance 3. Sustainable Business and Governance Practices o ESG integration in the C&M industry o Corporate governance and stakeholder accountability o Public trust and institutional integrity

List of Guided DSRG and DSRG*X* Research within the Digital Citizenship and Cyberwellness (DCC), Digital Inclusion (DI), Future of Work (FW), and Sustainability & Environment, Social, and Governance (SESG) Categories

No.	Code	Research Title	Research Category and Gap/Developmental Area
1	DCC-1	**The Content Code and the General Consumer Code: Assessing Their Awareness, Relevance, Usability, and Perceived Benefits Among the Malaysian Public in the Communications and Multimedia Industry**	Falling under the Guided Research Category of Digital Citizenship and Cyberwellness, this study aims to generate evidence that will aid in the validation and improvement of awareness and self-regulation of the identified industry codes on the Malaysian public.
2	DCC-2	**Evaluating *Kempen Internet Selamat*: Developing a Standardised Framework for Monitoring, Enhancing, and Measuring Online Safety Impact**	Falling under the Guided Research Category of Digital Citizenship and Cyberwellness, this study aims to generate evidence that will aid in the validation and improvement of awareness and self-regulation of *Kempen Internet Selamat* (KIS).
3	DCC-3	**Strengthening Public Adoption of Consumer Redress Mechanisms in the Communications, Multimedia, and Postal Services Industry: A Case Study in Negeri Sembilan**	Falling under the Guided Research Category of Digital Citizenship and Cyberwellness, this study aims to generate evidence that will aid in the awareness and adoption consumer redress mechanisms serving users of communications, multimedia and postal services industry.
4	DCC-4	**Cashless Payment Adoption in Negeri Sembilan: A Case Study of the General Public and Government Sector under the MCMC Cashless Society Scaling Up Programme**	Falling under the Guided Research Category of Digital Citizenship and Cyberwellness, this study aims to generate evidence on the MCMC Cashless Society Scaling Up Programme and examines general public and government sector readiness, challenges, and strategic opportunities to support inclusive digital transformation.

No.	Code	Research Title	Research Category and Gap/Developmental Area
5	DCC-5	**Understanding Digital Content Access Norms: Behavioural Archetypes and Ethical Perspectives Among Young Citizens**	Falling under the Guided Research Category of Digital Citizenship and Cyberwellness, this study aims to generate evidence related to digital content access norms amongst young citizens.
6	DI-1	**Investigating the Influence of Industrialised Building System (IBS) Precast Technology on Indoor Cellular Network Performance in Selangor Residential Developments**	Falling under the Guided Research Category of Digital Inclusion (DI), this study aims to generate evidence to contribute to the development of policies and guidelines to improve the quality of in-building cellular connectivity.
7	DI–2	**Assessing Langkawi's Digital Infrastructure for Inclusive Health, Assisted Living, and Wellness Tourism**	Falling under the Guided Research Category of Digital Inclusion (DI), this study aims to generate evidence to contribute to the development of programme evaluation, assessment and impact of efforts to improve access to health, assisted living, wellbeing and health tourism services.
8	DI-3	**Digital Skills Training at NADI: Assessing AI Tutors and Human Educators for Future Workforce Readiness**	Falling under the Guided Research Category of Digital Inclusion (DI), this study aims to generate evidence to address the digital divide and to ensure access to quality digital education and skills.
9	DI-4	**An Assessment of the Functionality and Connectivity of Master Antenna Television (MATV) Systems in Affordable High-Rise Residential Buildings in the Central Region**	Falling under the Guided Research Category of Digital Inclusion (DI), this study aims to generate evidence to address the digital divide by evaluating connectivity and its impact on access to broadcast information.
10	DI-5	**Aligning Connectivity and Development: Evaluating JENDELA's Impact in Rural Sabah Communities with Infrastructure Gaps**	Falling under the Guided Research Category of Digital Inclusion, this research aims to generate evidence on how JENDELA-enabled connectivity in rural Sabah influences socio-economic impact amid ongoing infrastructure gaps.

No.	Code	Research Title	Research Category and Gap/Developmental Area
11	DI-6	**Positioning 5G as a Public Utility in Sabah: Strategies for Inclusive and Impactful Connectivity**	Falling under the Guided Research Category of Digital Inclusion, this research aims to generate evidence to support equitable connectivity by identifying strategies and measures for 5G adoption in Sabah. The study focuses on addressing the digital divide, particularly among underserved communities, in ways that enable meaningful participation in digitalisation and society.
12	DI-7	**Enabling FTTH Adoption in Brownfield Zones: A Sabah Case Study on Infrastructure Diversity and Consumer Choice**	Falling under the Guided Research Category of Digital Inclusion, this research aims to generate evidence to support digital inclusion by identifying strategies and measures for FTTH adoption in Sabah. The study focuses on factors for ensuring equitable connectivity and addressing the digital divide, and to provide insights to ensure the sustainability on non-incumbent infrastructure providers.
13	FW-1	**Building a Sustainable Future of Work: Assessing Flexible Working Hours in the Sabah Telecommunications Sector**	Falling under the Guided Research Category of Future of Work, this research aims to generate evidence that will aid in the assessment of the adoption of Flexible Working Arrangements (FWA) within the telecommunications sector.
14	SESG-1	**Catalysing ESG Integration in the Communications and Multimedia Industry: MCMC's Role in Policy Development and Best Practice Facilitation**	Falling under the Guided Research Category of Sustainability & Environment, Social, and Governance (SESG), this study aims to generate evidence that will aid in the development of MCMC policies and facilitation of ESG adoption amongst C&M industry players.
15	SESG-2	**Transforming the Malaysian Communications and Multimedia Designated Industry Fora: A Governance and Operational Review**	Falling under the Guided Research Category of Sustainability & Environment, Social, and Governance (SESG), this study aims to generate evidence that will aid in the development of MCMC

No.	Code	Research Title	Research Category and Gap/Developmental Area
			policies supporting and facilitating the adoption of SESG adoption within the C&M designated industry fora.
16	SESG-3	**Enhancing the Financial Sustainability of Designated Industry Fora in Malaysia: Exploring Alternative Mechanisms for Long-Term Viability**	Falling under the Guided Research Category of Sustainability & Environment, Social, and Governance (SESG), this study aims to generate evidence that will aid in the development of relevant policies supporting and facilitating corporate governance and stakeholder accountability within the designated industry fora.
17	SESG-4	**A Study on Public Utility Perception and Legal Safeguards for Fibre Infrastructure in Sabah: Addressing Vandalism, Theft, and Community Awareness**	Falling under the Guided Research Category of Sustainability & Environment, Social, and Governance (SESG), this research contributes to the Sustainability and ESG agenda by examining governance mechanisms, community engagement, and legal safeguards that support the long-term resilience and public trust in Sabah's telecommunications infrastructure.
18	SESG-5	**SW110 Waste Management in Sabah: Bridging Infrastructure Gaps and Evaluating Localised Treatment Feasibility**	Falling under the Guided Research Category of Sustainability & Environment, Social, and Governance (SESG), this research contributes to the Sustainability and ESG agenda by critically examining governance mechanisms, policy frameworks, and legal provisions surrounding e-waste management in Sabah, highlighting gaps between regulation and real-world enforcement.
19	SESG-6	**Evaluating the Relevance of Regulatory Definitions in the General Consumer Code (GCC) Amidst Malaysia's Evolving Communications and Multimedia Landscape**	Falling under the Guided Research Category of Sustainability & Environment, Social, and Governance (SESG), this study aims to determine the relevance of the General Consumer Code of Practice for the Communications

No.	Code	Research Title	Research Category and Gap/Developmental Area
			and Multimedia Industry Malaysia (GCC) developed by the CFM in ensuring sustainable business and consumer practices through sound corporate governance and stakeholder accountability.
20	SESG-7	**Strengthening Business Stability in Malaysian Broadcasting through Sustainable Local Content Creation: A Strategic Framework for Producers and Regional Broadcasters**	Falling under the Guided Research Category of Sustainability & Environment, Social, and Governance (SESG), this study aims to understand the factors contributing to and impeding the sustainability of local Malaysian content creation towards ensuring the sustainability of Malaysia broadcasters and local content creators.
21	SESG-8	**Closing the Data Gaps: Developing Locally Representative Emission Factors for Malaysia's Transportation Sector to Strengthen Climate Policy and Reporting**	Falling under the Guided Research Category of Sustainability & Environment, Social, and Governance (SESG), this study contributes to the SESG agenda by examining environmental emission factors and measurement methodologies relevant to industries heavily dependent on transportation (e.g., logistics, delivery services), with a focus on the postal and courier sector's transition toward low-carbon fleet operations.

DCC-1 – The Content Code and the General Consumer Code: Assessing Their Awareness, Relevance, Usability, and Perceived Benefits Among the Malaysian Public in the Communications and Multimedia Industry

A. Research Field:

- Digital Citizenship and Cyberwellness (DCC) awareness and self-regulation.

B. Research Focus:

- Validation and improvement of designated industry fora voluntary industry codes.

C. Research Sponsor:

- Forum Strategy Planning Department, Forum Governance and Coordination Division, MCMC.

D. Contact Person for Queries:

- Ms. Elvira Shamsuddin, Director II, Forum Strategy Planning Department, Forum Governance and Coordination Division, elvira@mcmc.gov.my

E. Research Category:

- Digital Citizenship and Cyberwellness.
  This study aims to generate evidence that will aid in the validation and improvement of awareness and self-regulation of the identified industry codes on the Malaysian public.

F. Target Groups

1. Stakeholders Benefiting from the Research
The findings and recommendations derived from this research are anticipated to yield benefits for these stakeholder groups:

i. **Malaysian Communications and Multimedia Commission** (MCMC): To inform regulatory oversight, strategic planning and policy refinement related to the promotion and implementation of the Content Code and General Consumer Code.

ii. **Communications and Multimedia Content Forum of Malaysia** (CMCF): To strengthen public and industry engagement with the Content

Code and improve the relevance and accessibility of content regulation mechanisms.

iii. **Communications and Multimedia Consumer Forum of Malaysia** (CFM): To enhance outreach, education and support efforts aimed at increasing awareness and utilisation of the General Consumer Code among users.

iv. **General Public** (Communications and Multimedia Users): As beneficiaries of improved consumer protection and content standards, based on more accessible and better-utilised codes.

v. **Industry Players and Service Providers** in the Communications and Multimedia Sector: To gain clarity on compliance expectations and benefit from increased consumer confidence in self-regulation mechanisms.

2. Groups/Communities as Interviewees or Survey Respondents for the Research

To ensure a comprehensive and well-rounded perspective, the research will engage with a diverse range of stakeholders, who will constitute the primary interviewees or survey respondents:

i. **Representatives from CMCF**: To provide insights into current challenges and strategies related to code promotion and industry adoption.

ii. **Representatives from CFM**: To share perspectives on public education efforts, user engagement and the practical application of the General Consumer Code.

iii. **Adult Users of Communications and Multimedia Services** in Malaysia: To assess public awareness, understanding and utilisation of the Content and General Consumer Codes.

iv. **Consumers with Direct Experience in Service or Content-Related Issues**: To gather real-world insights into how individuals perceive, engage with, or seek recourse through the codes.

(Where appropriate, researchers are encouraged to propose refinements to respondent categories to better align with the research aims.)

RESEARCH PROBLEM/CONTEXT

The Communications and Multimedia Act (CMA) provides for the establishment of industry fora and the development of consumer codes to promote self-regulation within the sector. Among these, the two primary codes that directly affect consumers—the Content Code[3] and the General Consumer Code[4]—were first introduced and registered in 2001. These Codes were designed to strengthen consumer protection and enhance content regulation in line with the CMA's objectives.

Both Codes have since been updated, with the most recent versions released in 2022, reflecting evolving industry practices and regulatory needs. However, despite being legally mandated and in force for over two decades, the Codes appear to be underutilised by both industry players and the general public. This limited engagement raises important questions about their practical relevance, visibility, and accessibility. Given their intended role in upholding consumer rights and promoting responsible content standards, it is essential to assess how effectively these Codes serve the public—particularly in the areas of consumer protection and content regulation.

RESEARCH AIMS

The Forum Governance and Coordination Division (FGCD) is responsible for ensuring that industry fora operate effectively and align with the broader objective of protecting and educating the public. This research aims to provide actionable insights to FGCD, the Malaysian Communications and Multimedia Commission (MCMC), and the industry fora to evaluate and enhance the effectiveness of the Content Code and General Consumer Code in supporting the public in managing services provided by licensees in Malaysia.

Ultimately, the study seeks to strengthen the self-regulatory ecosystem by promoting greater consumer protection and ensuring that services and content delivered within the communications and multimedia industry reflect national aspirations and serve the public interest.

[3] Content Code: Communications and Multimedia Content Forum of Malaysia. (2022). Malaysian Communications and Multimedia Content Code 2022. Malaysian Communications and Multimedia Commission. Accessed on 14 July 2025 at <https://www.mcmc.gov.my/skmmgovmy/media/General/pdf/Content-Code-2022.pdf>

[4] General Consumer Code: Communications and Multimedia Consumer Forum of Malaysia. (2022). General Consumer Code of Practice for the Communications and Multimedia Industry Malaysia (2nd ed.). Malaysian Communications and Multimedia Commission. Accessed on 14 July, 2025 at <https://www.mcmc.gov.my/skmmgovmy/media/General/registers/cma/General_Consumer_Code_of_Practice_Second_Edition_Registered.pdf>

RESEARCH OBJECTIVES

Researchers are invited to submit proposals guided by the following overarching Research Objectives (ROs) and Research Questions (RQs):

RO 1: To assess the accessibility and effectiveness of the Content Code and General Consumer Code in fulfilling the objectives of the Communications and Multimedia Act (CMA) and protecting consumer rights.

- RQ 1.1: How accessible are the Content Code and General Consumer Code to the public in terms of format, language, and dissemination?
- RQ 1.2: To what extent do these Codes align with the consumer protection goals outlined in the CMA?
- RQ 1.3: How effective are the Codes in facilitating the resolution of consumer complaints related to service delivery and content regulation?

RO 2: To evaluate the level of public awareness and understanding of the Content Code and General Consumer Code within Malaysia's communications and multimedia industry

- RQ 2.1: What proportion of the public is aware of the existence and purpose of the Content Code and General Consumer Code?
- RQ 2.2: What is the level of public understanding regarding the provisions and functions of these Codes?
- RQ 2.3: Through which channels do consumers typically learn about these Codes?

RO 3: To examine how, and under what circumstances, members of the public engage with or utilise the Codes in addressing service-related or content-related issues.

- RQ 3.1: In what situations do consumers refer to or rely on the Codes to address service or content-related concerns?
- RQ 3.2: What barriers hinder public engagement with the Codes when facing such issues?
- RQ 3.3 What role do service providers play in facilitating consumer use of the Codes?

RO 4: To explore the level of public trust in service providers and in the roles of CMCF and CFM Forum as self-regulatory bodies.

- RQ 4.1: To what extent do consumers trust service providers to comply with the Codes?
- RQ 4.2: How is the effectiveness and impartiality of the CMCF and CFM perceived by the public?
- RQ 4.3: How does public trust in these bodies influence consumer engagement with the Codes?

RO 5: To propose improvements that enhance the relevance, usability, and overall effectiveness of the Codes in serving the public interest.

- RQ 5.1: What strategies can improve the visibility and accessibility of the Codes?
- RQ 5.2: How can the content or structure of the Codes be refined to better reflect current consumer needs?
- RQ 5.3: What initiatives can be introduced to encourage greater public
- engagement and utilisation of the Codes?

(The researcher is invited to propose refinements to Research Objectives and Research Questions. However, the researcher is requested to provide an explanation of the anticipated value of the proposed changes and potential benefits of proposed refinements.)

DCC-2 – Evaluating Kempen Internet Selamat: Developing a Standardised Framework for Monitoring, Enhancing, and Measuring Online Safety Impact

A. Research Field:

- Digital Citizenship, Online Safety and Media Literacy.

B. Research Focus:

- The focus of the research is on the programme evaluation, assessment impact and measurement of online safety and media literacy initiatives.

C. Research Sponsor:

- Community Engagement Department, Regional and State Coordination Division, MCMC.

D. Contact Person for Queries:

- Ms. Nurul Amira Mat Ariffin, Deputy Director, Community Engagement Department, Regional and State Coordination Division, amira.ariffin@mcmc.gov.my

E. Research Category:

- Digital Citizenship and Cyberwellness (DCC).
  This study aims to generate evidence that will aid in the validation and improvement of awareness and self-regulation of *Kempen Internet Selamat* (KIS).

F. Target Groups

1. Stakeholders Benefiting from the Research
The findings and recommendations derived from this research are anticipated to yield benefits for these stakeholder groups:

i. **Government and Regulators** responsible for policy and programme design (framework adoption, policy alignment, integration into national digital literacy strategies, regional benchmarking) including:

- Ministry of Communications
- Ministry of Education
- Malaysian Communications and Multimedia Commission (MCMC)
- Regional and international partners (ASEAN, ITU)

- Policy Makers and Regulators

ii. **Programme Providers** responsible for implementation and collaboration (training enhancements, delivery model optimisation, cross sector collaboration, communication strategy refinement) including:

- National Dissemination of Information Centres (NADI)
- Programme facilitators and trainers
- Online service providers and telcos
- Media and communication teams

2. Groups/Communities as Interviewees or Survey Respondents for the Research

To ensure a comprehensive and well-rounded perspective, the research will engage with a diverse range of stakeholders, who will constitute the primary interviewees or survey respondents:

i. **Programme Providers** involved in the design, delivery and evaluation (feedback, informed module design, improved delivery formats, evaluation tool usability) including:

- KIS Facilitators and Trainers
- Media and communications personnel
- School administrators and educators

ii. **Programme Users and Extended Stakeholders** (content relevance, behavioural impact insights, demographic specific engagement strategies) including:

- Students (school and university)
- Parents and households University Lecturers and Students
- Community members
- The general public (segmented by age, region, digital literacy level or other relevant criteria)

(Where appropriate, researchers are encouraged to propose refinements to respondent categories to better align with the research aims.)

RESEARCH PROBLEM/CONTEXT

Kempen Internet Selamat (KIS) is a national initiative spearheaded by the Ministry of Communications and the Malaysian Communications and Multimedia Commission (MCMC) to raise public awareness on online safety. It addresses pressing digital threats such as online scams, cyberbullying, misinformation, online gambling, child sexual abuse material (CSAM), and age-inappropriate content on social media.

KIS is delivered through structured educational modules tailored to different age groups and implemented across schools, higher education institutions, and community settings.

Despite its wide reach—having engaged 221 schools nationwide as of July 2025—KIS currently lacks a standardised Monitoring and Evaluation (M&E) framework. Its effectiveness is primarily measured through output indicators (e.g., number of events or participants), rather than outcome-based metrics such as knowledge retention, behavioural change, or long-term community impact. Additionally, key campaign components like media promotions, branding, and community events are not systematically evaluated, leaving their influence on public engagement largely anecdotal.

This gap limits the ability of programme implementers and policymakers to assess the true impact of KIS across different demographics and delivery formats. Without validated tools and indicators, it is difficult to ensure that the campaign remains relevant, impactful, and evidence-driven.

The proposed research will deliver a validated and adaptable M&E framework, enabling more accurate and meaningful assessment of KIS's effectiveness. This will allow the Community Engagement Division (CED) to transition from output-based reporting to outcome-focused evaluation—supporting continuous improvement, strategic decision-making, and alignment with national digital safety goals.

This research will contribute to a more evidence-based, impactful, and sustainable implementation of KIS, aligned with MCMC's mission to protect and empower the public in the digital landscape.

RESEARCH AIMS

To develop a comprehensive and standardised Monitoring and Evaluation (M&E) framework for *Kempen Internet Selamat* (KIS) that effectively measures the campaign's impact across its core implementation areas—educational programmes, media promotions, events, and branding efforts. The framework will be designed for practical use, enabling consistent monitoring, strategic

enhancement, and outcome-based assessment of online safety initiatives. By emphasising scalability and standardisation, the research aims to support broader adoption and institutional integration across national digital literacy and safety programmes.

RESEARCH OBJECTIVES

Researchers are invited to submit proposals guided by the following overarching Research Objectives (ROs) and Research Questions (RQs):

RO 1: To conduct a comparative study of selected Monitoring and Evaluation (M&E) frameworks from local and international Internet safety awareness programmes, identifying best practices and relevant components adaptable to the Malaysian context.

- RQ 1.1: What are the core components and indicators used in existing M&E frameworks for online safety awareness campaigns?
- RQ 1.2: How do these frameworks measure behavioural change, knowledge retention, and engagement?
- RQ 1.3: Which elements are most adaptable to the Malaysian context, particularly for *Kempen Internet Selamat* (KIS)?

RO 2: To assess the current implementation of Internet safety awareness modules in Malaysia, with a focus on identifying gaps in impact measurement and evaluation practices and exploring opportunities to transition from output-based to outcome-based metrics.

- RQ 2.1: What evaluation methods are currently used in KIS to measure programme effectiveness?
- RQ 2.2: What are the limitations of current output-based metrics in capturing behavioural and long-term impact?
- RQ 2.3: How do stakeholders perceive the effectiveness of KIS across different delivery formats and demographic segments?

RO 3: To design a Monitoring and Evaluation framework that can be applied across various programme formats and delivery methods within Internet safety awareness efforts.

- RQ 3.1: What indicators and tools can be developed to measure the effectiveness of KIS across educational, media, and community engagement components?

- RQ 3.2: How can the framework accommodate diverse delivery methods and demographic profiles?

- RQ 3.3: What validation processes are needed to ensure the reliability, usability, and scalability of the proposed framework?

(The researcher is invited to propose refinements to Research Objectives and Research Questions. However, the researcher is requested to provide an explanation of the anticipated value of the proposed changes and potential benefits of proposed refinements.)

DCC-3 – Strengthening Public Adoption of Consumer Redress Mechanisms in the Communications, Multimedia, and Postal Services Industry: A Case Study in Negeri Sembilan

A. Research Field:

- Digital Citizenship, Consumer Protection.

B. Research Focus:

- The focus of the research is on the awareness and adoption of consumer redress mechanisms.

C. Research Sponsor:

- Negeri Sembilan State Office, Southern Regional Office, MCMC.

D. Contact Person for Queries:

- Ts. Ahmad Faizal Rosehaizad, Section Head (Regulatory), Negeri Sembilan State Office, Southern Regional Office, faizal.rosehaizad@mcmc.gov.my

E. Research Category:

- Digital Citizenship and Cyberwellness (DCC).
  This study aims to generate evidence that will aid the awareness and adoption of consumer redress mechanisms serving users of communications, multimedia and postal services industry.

F. Target Groups

1. Stakeholders Benefiting from the Research
The findings and recommendations derived from this research are anticipated to yield benefits for these stakeholder groups:

i. **Consumers in Negeri Sembilan**, particularly those who use telecommunications services and may face service-related issues.

ii. **MCMC and related regulatory bodies**, who will gain insights into public awareness gaps and receive recommendations to improve outreach and engagement strategies.

iii. **Local community leaders and digital literacy advocates**, who can use the findings to support consumer education and empowerment initiatives.

2. Groups/Communities as Interviewees or Survey Respondents for the Research

To ensure a comprehensive and well-rounded perspective, the research will engage with a diverse range of stakeholders, who will constitute the primary interviewees or survey respondents:

i. **General public in Negeri Sembilan**, including individuals from diverse age groups, income levels, and geographic areas (urban and rural), to assess awareness and usage of the Complaints Redress Mechanism and Escalation Process.

ii. **Telecommunications service users**, who have experienced service issues, to understand their complaint behaviour and familiarity with the portal.

iii. **Representatives from the designated industry fora**, Communications and Multimedia Consumer Forum of Malaysia (CFM), Communications and Multimedia Content Forum of Malaysia (CMCF) and Postal Forum).

iv. **Community representatives and local digital outreach personnel**, to gather perspectives on current communication efforts and challenges in promoting the portal.

(Where appropriate, researchers are encouraged to propose refinements to respondent categories to better align with the research aims.)

RESEARCH PROBLEM/CONTEXT

The Malaysian Communications and Multimedia Commission (MCMC) envisions a communications, multimedia, and postal services industry that is competitive, efficient, and increasingly self-regulating — one that supports Malaysia's economic and social development. Central to this vision is the empowerment of consumers through access to reliable services, transparent redress mechanisms, and informed participation in the digital ecosystem.

Under the Communications and Multimedia Act 1998, MCMC is mandated to uphold national policy objectives that include promoting consumer interests, ensuring service quality, and facilitating fair and effective dispute resolution. A key component of this mandate is the structured complaint escalation pathway, which begins with service providers and progresses through focal officers and designated

industry fora — such as the Communications and Multimedia Consumer Forum of Malaysia (CFM), the Communications and Multimedia Content Forum of Malaysia (CMCF), and the Postal Forum — culminating in the MCMC Consumer Redress Portal (*Aduan MCMC*) as the final escalation point when prior avenues have been exhausted.

In Negeri Sembilan, preliminary observations suggest that public awareness of this multi-tiered redress framework remains limited. Many consumers are unaware of the intermediary steps or mistakenly perceive *Aduan MCMC* as the initial point of contact. This lack of clarity can lead to misdirected complaints, delayed resolutions, and reduced confidence in regulatory processes.

This research seeks to examine the extent of public understanding of the full complaint escalation pathway across the communications, multimedia, and postal services sectors. It aims to identify barriers to effective engagement and propose strategies to enhance visibility, accessibility, and utilisation of all redress mechanisms. In doing so, the study supports MCMC's mission to promote consumer empowerment, uphold service standards, and foster collaborative and transparent regulatory practices.

RESEARCH AIMS

To investigate public awareness, understanding, and utilisation of the full complaint escalation pathway for communications, multimedia, and postal services in Negeri Sembilan — encompassing service providers, focal officers, designated industry fora, and culminating in the MCMC Consumer Redress Portal (*Aduan MCMC*) — and to develop actionable recommendations for enhancing consumer engagement and improving the effectiveness of complaint resolution mechanisms.

RESEARCH OBJECTIVES

Researchers are invited to submit proposals guided by the following overarching Research Objectives (ROs) and Research Questions (RQs):

RO 1: To assess the level of public awareness and understanding of each stage in the complaint escalation process.

- RQ 1.1: What do consumers in Negeri Sembilan know about the complaint escalation process in the telecommunications sector?
- RQ 1.2: How familiar are they with the roles of focal officers and designated fora?

RO 2: To identify barriers and motivations influencing consumer engagement with focal officers, designated fora, and the *Aduan MCMC* portal.

- RQ 2.1: What factors influence consumer decisions to escalate complaints beyond service providers?

- RQ 2.2: What challenges do consumers face when attempting to use the Aduan *MCMC* portal?

RO 3: To evaluate the effectiveness of current outreach and communication strategies related to consumer redress mechanisms.

- RQ 3.1: How effective are current communication efforts in promoting awareness of redress mechanisms?

RO 4: To propose targeted communication and education strategies to enhance visibility and usage of the full complaint pathway.

- RQ 4.1: What improvements can be made to streamline and clarify the complaint pathway for consumers?

(The researcher is invited to propose refinements to Research Objectives and Research Questions. However, the researcher is requested to provide an explanation of the anticipated value of the proposed changes and potential benefits of proposed refinements.)

DCC-4 – Cashless Payment Adoption in Negeri Sembilan: A Case Study of the General Public and Government Sector under the MCMC Cashless Society Scaling Up Programme

A. Research Field:

- Digital Citizenship, Inclusive Digital Transformation.

B. Research Focus:

- The focus of the research is on the awareness and adoption of the MCMC Cashless Society Scaling Up Programme.

C. Research Sponsor:

- Negeri Sembilan State Office, Southern Regional Office, MCMC.

D. Contact Person for Queries:

- Ms. Ema Shuaida Shuaid, Section Head (Development), Negeri Sembilan State Office, Southern Regional Office, shuaida.shuaid@mcmc.gov.my

E. Research Category:

- Digital Citizenship and Cyberwellness (DCC).
  This study aims to generate evidence on the MCMC Cashless Society Scaling Up Programme and examines general public and government sector readiness, challenges, and strategic opportunities to support inclusive digital transformation.

F. Target Groups

1. **Stakeholders Benefiting from the Research**
   The findings and recommendations derived from this research are anticipated to yield benefits for these stakeholder groups:

   i. **Consumers in Negeri Sembilan**: Improved access to cashless payment options, greater convenience, and enhanced financial inclusion, especially in areas targeted by the MCMC Cashless Society Scaling Up Programme.

   ii. **Small Traders and Retailers**: Strategic insights and practical recommendations to help them adopt digital payment systems and expand their market reach through digital platforms, in line with the merchant enablement pillar of the programme.

iii. **Local Government Agencies and Policymakers**: Evidence-based data to guide policy formulation, infrastructure planning, and digital economy initiatives, supporting the public sector integration goals of the MCMC Cashless Society Scaling Up Programme.

iv. **MCMC and Related Stakeholders**: Support for national digital transformation goals and alignment with the MCMC Cashless Society Scaling Up Programme's five strategic pillars and DSRG objectives.

Note: While the research is anchored in the MCMC Cashless Society Scaling Up (Cashless Society) Programme, it also seeks to uncover additional insights and community-driven practices that may inform future iterations or complementary initiatives.

2. Groups/Communities as Interviewees or Survey Respondents for the Research

To ensure a comprehensive and well-rounded perspective, the research will engage with a diverse range of stakeholders, who will constitute the primary interviewees or survey respondents:

i. **General public in Negeri Sembilan**, including various age groups, income levels, and geographic locations (urban and rural), to understand daily usage patterns, barriers, and perceptions of cashless systems promoted under the Cashless Society Programme.

ii. **Small traders and micro-retailers**, such as hawkers, market vendors, and small shop owners, to explore their experiences, challenges, and aspirations related to cashless adoption, particularly those engaged or eligible under the programme's merchant onboarding efforts.

iii. **Local government officials and digital economy programme coordinators**, to gather insights on current initiatives, infrastructure readiness, and policy support, with emphasis on the implementation of the Cashless Society Programme at the state and municipal levels.

(Where appropriate, researchers are encouraged to propose refinements to respondent categories to better align with the research aims.)

RESEARCH PROBLEM/CONTEXT

The transition toward a cashless society is a cornerstone of Malaysia's digital transformation agenda, supported by national frameworks such as MyDIGITAL, the Negeri Sembilan Digital Economy Blueprint 2027, and most recently, the MCMC Cashless Society Scaling Up Programme launched in 2024[5]. This programme represents a strategic push to accelerate digital payment adoption across underserved communities, micro-enterprises, and public sector services. Structured around five strategic pillars, the MCMC Cashless Society Scaling Up Programme addresses the ecosystem required for sustainable cashless adoption.

The first pillar, **Infrastructure Readiness**, focuses on expanding internet connectivity and ensuring the availability of digital payment terminals, particularly in semi-urban and rural areas.

The second pillar, **Digital Literacy and Awareness**, aims to equip citizens and merchants with the knowledge and confidence to use digital payment platforms safely and effectively. This includes support for community-based outreach and education initiatives such as the *Kempen Internet Selamat*, which promotes responsible and informed digital behaviour.

The third pillar, **Merchant Onboarding and Enablement**, provides targeted support for small traders and micro-retailers to adopt digital payment systems, including technical assistance, simplified registration processes, and partnerships with fintech providers.

The fourth pillar, **Public Sector Integration**, promotes the digitisation of government services—such as licensing, tax collection, and utility payments—by encouraging local authorities to embed cashless systems into their operations.

Finally, the fifth pillar, **Monitoring and Impact Evaluation**, establishes mechanisms to track adoption metrics, user satisfaction, and programme effectiveness, enabling data-driven policy refinement.

Despite these efforts, the pace and depth of adoption in Negeri Sembilan remain uneven. Challenges such as institutional readiness, fragmented policy execution, limited public trust, and infrastructural gaps continue to hinder full-scale implementation. This research proposes a case study approach to examine the adoption of cashless payment systems in Negeri Sembilan, focusing on two distinct segments: the general public (including small traders and consumers) and the government sector (including public service delivery and administrative functions). By exploring adoption patterns, barriers, and enablers across these segments, the study aims to generate actionable insights that support the goals of the MCMC

[5] BERNAMA. (2024, July 3). *MCMC launches Cashless Society Scaling Up Programme to boost digital payment adoption*. Accessed on 24 July 2025 at <https://www.bernama.com/en/news.php?id=2385658>

Cashless Society Scaling Up Programme and inform localised strategies for inclusive digital transformation.

While this study is anchored in the MCMC Cashless Society Scaling Up Programme, it also seeks to uncover additional insights, community-driven practices, and contextual factors that may inform future iterations or complementary digital payment initiatives. This approach ensures that the research remains responsive to local realities while contributing to broader national goals.

RESEARCH AIMS

This study aims to examine the adoption of cashless payment systems in Negeri Sembilan by assessing current usage patterns, identifying key enablers and barriers, and evaluating the impact of national and state-level initiatives such as the MCMC Cashless Society Scaling Up Programme. By focusing on both the general public (including small traders and retailers) and the government sector, the research will generate evidence-based insights to inform policy development, strategic planning, and digital inclusion efforts. Ultimately, the study seeks to propose practical recommendations that support inclusive and effective digital transformation across public services and administrative operations.

RESEARCH OBJECTIVES

Researchers are invited to submit proposals guided by the following overarching Research Objectives (ROs) and Research Questions (RQs):

RO 1: To evaluate the current adoption rate and usage patterns of cashless payment systems among the general public and government sector in Negeri Sembilan, with specific reference to the implementation and outreach of the MCMC Cashless Society Scaling Up Programme.

- RQ 1.1: What is the current level of cashless payment usage among the general public and government agencies in Negeri Sembilan under the MCMC Cashless Society Scaling Up Programme?

- RQ 1.2: What types of digital payment platforms are most used in each segment?

- RQ 1.3: What demographic or institutional factors influence adoption rates?

RO 2: To identify and analyse the technological, organisational, and societal factors influencing adoption, using surveys, interviews, and policy reviews, with a focus on the enablers and barriers addressed by the MCMC Cashless Society Scaling Up Programme's five strategic pillars.

- RQ 2.1: What technological infrastructure supported by the programme facilitates or limits adoption in each segment?

- RQ 2.2: How do organisational policies and leadership within government services align with the programme's objectives?

- RQ 2.3: What societal attitudes, trust levels, and behavioural norms affect public adoption, and how are these being addressed by the programme?

RO 3: To assess the digital readiness and challenges faced by small traders and retailers, including infrastructure, literacy, and customer demand, in relation to the merchant enablement and digital literacy components of the MCMC Cashless Society Scaling Up Programme.

- RQ 3.1: What is the level of digital literacy among small traders and retailers targeted by the programme?

- RQ 3.2: What infrastructural gaps (e.g., connectivity, devices) remain despite programme interventions?

- RQ 3.3: How does customer demand influence traders' willingness to adopt cashless systems, and how is this being shaped by the programme's outreach?

RO 4: To develop strategic recommendations and implementation guidelines based on empirical findings and best practices, aligned with the goals and evaluation framework of MCMC Cashless Society Scaling Up Programme.

- RQ 4.1: What lessons can be learnt from successful adopters engaged through the programme?

- RQ 4.2: What policy or programme interventions could further accelerate adoption in underserved areas beyond current programme scope?

- RQ 4.3: How can digital inclusion be strengthened to ensure equitable access to cashless systems in line with the programme's national objectives?

(The researcher is invited to propose refinements to Research Objectives and Research Questions. However, the researcher is requested to provide an explanation of the anticipated value of the proposed changes and potential benefits of proposed refinements.)

DCC-5 – Understanding Digital Content Access Norms: Behavioural Archetypes and Ethical Perspectives Among Young Citizens

A. Research Field:

- Digital Citizenship.

B. Research Focus:

- Young citizens' digital content access norms.

C. Research Sponsor:

- Astro Malaysia Holdings Berhad.

D. Contact Persons for Queries:

- Ms. Eva Piramila Poovan, VP, Head Group Data and Revenue Assurance, Group Data Division, Astro Malaysia Holdings Berhad, evapiramila_poovan@astro.com.my
- Mr. Farquar Haqqani Fadhlullah Suhaimi, Senior AVP Strategy, Group Strategy Division, Astro Malaysia Holdings Berhad, farquar_haqqani@astro.com.my

E. Research Category:

- Digital Citizenship and Cyberwellness.
  This study aims to generate evidence related to digital content access norms amongst young citizens.

F. Target Groups

1. **Stakeholders Benefiting from the Research**
   The findings and recommendations derived from this research are anticipated to yield benefits for these stakeholder groups:

   i. **Broadcasters (including subscription-based providers)** to inform content strategy, pricing models, customer education, and targeted anti-piracy communications.

   ii. **Policy-makers and Regulators** such as MCMC and other relevant agencies, who can use the findings to shape youth-centric digital content policies, enforcement strategies, and public education efforts.

iii. **Educational Institutions** including secondary schools, universities, and digital literacy programmes, which can integrate findings into curricula and awareness campaigns focused on ethical digital citizenship.

iv. **Content Creators and IP holders** to better understand youth consumption patterns and ethical reasoning, enabling more effective engagement, protection strategies, and advocacy for fair content usage.

v. **Youth Advocacy and CSR Organisations** promoting responsible media consumption and ethical digital engagement among young citizens.

2. Groups/Communities as Interviewees or Survey Respondents for the Research

To ensure a comprehensive and well-rounded perspective, the research will engage with a diverse range of stakeholders, who will constitute the primary interviewees or survey respondents:

i. **Age group: 16-17 (Minors)**

- Profile: Secondary school students in urban and suburban settings and who are active digital content consumers.
- Characteristics: Digitally literate, frequent internet users, exposed to multiple content platforms including streaming, social media, and informal sharing channels.
- Ethical Considerations:
  - **Parental/Guardian consent** is required for participation.
  - **Youth assent** must be obtained, ensuring they understand the study, and agree voluntarily.
  - **Sensitive framing** of questions is essential to avoid inducing guilt or discomfort.
  - **Privacy protections** must be robust, with strict anonymity and data handling protocols.
  - **Institutional ethics** approval is recommended based on the research setting.

ii. **Age group: 18-22 (young adults in tertiary education or early career)**

- Profile: University students and early career professionals; highly active in digital ecosystems.
- Characteristics: High digital literacy, frequent engagement with both legal and informal content access methods, including subscription sharing and pirated streaming.

- Ethical Considerations:
  - **Informed consent** must be clearly communicated and documented.
  - **Anonymity and confidentiality** are critical due to the sensitivity of the topic.
  - **Voluntary participation** must be emphasised with no coercion or implied obligation.

iii. **Age Group: 23-30 (Young adults with evolving digital habits)**

- Profile: Early career professionals and young adults with families; digital content consumption may include household-level decisions.
- Characteristics: Exposure to a mix of legal and informal access channels; may engage in content sharing across family units or use unauthorized devices.
- Ethical Considerations:
  - Same as above (18-22), with additional sensitivity to family-related digital norms and shared access behaviours.
  - Ensure participants feel safe discussing household practices without judgement.

iv. **Additional Youth Segments (to enhance inclusivity)**

- Youth from rural or underserved communities to capture digital divide nuances.
- Youth with disabilities to ensure accessibility and inclusion.
- Youth from marginalised or underrepresented groups to reflect diverse digital experiences.

(Where appropriate, researchers are encouraged to propose refinements to respondent categories to better align with the research aims.)

RESEARCH PROBLEM/CONTEXT

Digital content consumption has become an integral part of everyday life for young citizens, shaping how they engage with entertainment, education, and social interaction. Within this landscape, a wide spectrum of access behaviours has emerged—ranging from fully legal and paid subscriptions to informal and unauthorised methods. Despite the increasing availability of legitimate content through affordable platforms, the consumption of pirated content remains prevalent, particularly among youth. This persistence suggests that traditional enforcement and deterrence strategies may not be addressing the underlying motivations and perceptions that drive such behaviours.

Pirated content is distributed through a variety of channels that are easily accessible and often normalised within peer communities. These include illegal streaming websites, peer-to-peer file sharing networks, social media platforms, messaging apps, and unauthorised IPTV or Android TV boxes. Additionally, cloud storage links and community fora often serve as informal repositories for pirated media. These channels offer convenience and breadth of content, often bypassing paywalls or geographic restrictions, which can make them appealing to users who may not fully perceive the ethical or legal implications of their choices.

At the same time, many young digital citizens engage in hybrid or alternative access practices that fall into a grey area of legitimacy. These include sharing paid subscriptions across households or friend groups, using free trials or ad-supported platforms, accessing content through educational or institutional licenses, and consuming user-generated or remixed content on platforms like YouTube or TikTok. Such behaviours reflect a complex interplay of digital literacy, socio-economic factors, platform design, and evolving ethical worldviews.

Understanding these diverse access norms is critical to developing more effective and empathetic strategies for promoting ethical digital engagement. Rather than assuming piracy is a foregone conclusion or a purely moral failing, this research seeks to explore how young people rationalise their choices, what values and perceptions shape their behaviours, and how these cluster into distinct behavioural archetypes. By elevating the perspectives of young digital citizens, the study aims to inform targeted interventions that reduce reliance on pirated content while respecting the lived realities and motivations of the target group.

RESEARCH AIMS

This study aims to explore and classify the behavioural archetypes of young digital citizens in relation to digital content access—including practices commonly referred to as piracy. By examining their motivations, ethical perspectives, and awareness of legal frameworks, the research seeks to generate insights that can inform targeted strategies to reduce reliance on unauthorised content access and promote ethical digital engagement.

RESEARCH OBJECTIVES

Researchers are invited to submit proposals guided by the following overarching Research Objectives (ROs) and Research Questions (RQs):

RO 1: To identify the common pathways through which young citizens access digital content.

- RQ 1.1: What factors (e.g., cost, convenience, platform availability) influence the choice of access method?
- RQ 1.2: How do users perceive the legitimacy or acceptability of different access methods?

RO 2: To explore the ethical perspectives and perceived harms associated with unauthorised content access.

- RQ 2.1: How do young digital citizens define and interpret the ethics of digital content access?
- RQ 2.2: What are their views on the impact of unauthorised access on creators, platforms, and society?
- RQ 2.3: How do these ethical perspectives vary across different demographic or behavioural groups?

RO 3: To assess awareness and understanding of legal frameworks and consequences related to digital content access.

- RQ 3.1: What is the level of awareness among youth regarding the legality of various content access practices?
- RQ 3.2: How do perceptions of enforcement and consequences influence behaviour?
- RQ 3.3: What sources of information shape their understanding of digital content laws?

RO 4: To develop behavioural archetypes based on motivations, ethical reasoning, and access patterns.

- RQ 4.1: What distinct behavioural profiles can be identified among young digital citizens?

- RQ 4.2: How do these archetypes differ in terms of motivations, ethical views, and awareness?

- RQ 4.3: What contextual factors (e.g., socio-demographics, digital literacy) influence archetype formation?

RO 5: To propose targeted engagement and policy strategies to reduce reliance on unauthorised content access.

- RQ 5.1: What types of messaging and interventions resonate with each behavioural archetype?

- RQ 5.2: How can policy, education, and platform design support ethical content consumption?

- RQ 5.3: What role can public-private partnerships play in shifting digital access norms?

(The researcher is invited to propose refinements to Research Objectives and Research Questions. However, the researcher is requested to provide an explanation of the anticipated value of the proposed changes and potential benefits of proposed refinements.)

DI-1 – Investigating the Influence of Industrialised Building System (IBS) Precast Technology on Indoor Cellular Network Performance in Selangor Residential Developments

A. Research Field:

- Malaysian Communications and Multimedia Network Coverage and Digital Inclusion.

B. Research Focus:

- MCMC Policy and Guideline Development related to Indoor Cellular Network Performance.

C. Research Sponsors:

- Selangor State Office, Central Regional Office, MCMC.
- Quality Monitoring Department, Quality Management Division, MCMC.

D. Contact Person for Queries:

- Mr. Muhammad Jaflus Baharudi, Section Head (Development), Selangor State Office, muhammad.jaflus@mcmc.gov.my
- Mr. Dean Ashraf Mohamed Zin, Director II, Quality Monitoring Department, Quality Management Division, ashraf.mdzin@mcmc.gov.my

E. Research Category:

- Digital Inclusion (DI).
  This study aims to generate evidence to contribute to the development of policies and guidelines to improve the quality of in-building cellular connectivity.

F. Target Groups

1. Stakeholders Benefiting from the Research

The findings and recommendations derived from this research are anticipated to yield benefits for these stakeholder groups:

i. **Members of the public** comprising existing and potential homeowners and residents.

ii. **Property developers in Selangor**, especially those employing IBS construction materials.

iii. **Cellular Network Providers** deploying or planning to deploy cellular towers in new residential areas.

iv. **Malaysian Communications and Multimedia Commission** to obtain evidence to support policy and guideline development.

2. Groups/Communities as Interviewees or Survey Respondents for the Research

To ensure a comprehensive and well-rounded perspective, the research will engage with a diverse range of stakeholders, who will constitute the primary interviewees or survey respondents:

i. **Members of the public** comprising existing and potential homeowners and residents residing in Selangor.

ii. **Selangor property developers**, especially those employing IBS construction materials.

iii. **Cellular companies** which have or will deploy cellular towers in new residential areas in Selangor.

(Where appropriate, researchers are encouraged to propose refinements to respondent categories to better align with the research aims.)

RESEARCH PROBLEM/CONTEXT

The Industrialised Building System (IBS), particularly precast concrete technology, is increasingly adopted in Malaysian residential developments due to its cost-efficiency, speed of construction, and quality control. However, recent investigations by MCMC have revealed a recurring issue: poor indoor cellular signal coverage in homes constructed using IBS precast components. Despite strong outdoor signal strength, indoor environments often record little to no connectivity.

This phenomenon is attributed to the high signal attenuation caused by dense concrete and embedded reinforcements in IBS structures, which significantly hinder radio frequency (RF) signal penetration. As Malaysia advances toward a digitally inclusive society, this issue poses a serious challenge to consumer protection, quality of service (QoS) compliance, and universal connectivity goals.

The research seeks to explore not only the extent of this issue but also to identify suitable cell radio planning approaches, particularly in terms of serving cell radius and its impact on downlink and uplink transmission performance. The study will include a comparative analysis with areas built using conventional construction methods, with the aim of proposing planning strategies that can effectively mitigate the impact of IBS on mobile network performance. These insights will inform policy and regulatory strategies, network planning, and collaborative frameworks between developers, telcos, and regulators.

RESEARCH AIMS

The research aims to investigate the impact of IBS precast construction on indoor cellular signal performance in Selangor residential areas, and to deduce optimal base station coverage strategies and planning parameters that ensure effective connectivity in such environments.

RESEARCH OBJECTIVES

Researchers are invited to submit proposals guided by the following overarching Research Objectives (ROs) and Research Questions (RQs):

RO 1: To assess the impact of IBS precast building materials on indoor cellular signal attenuation through field measurements or survey questionnaires and comparative analysis with non-IBS structures.

- RQ 1.1: To what extent do IBS precast materials attenuate indoor cellular signals compared to conventional building materials?

- RQ 1.2: What are the key material and structural characteristics of IBS that contribute to signal degradation?

RO 2: To forecast the 10-year adoption trend of Industrialised Building System (IBS) techniques in Malaysia by analysing publicly available adoption data, conducting surveys or interviews with selected developers, and assessing the potential influence of relevant government policies and incentives.

- RQ 2.1: What are the current patterns and influencing factors in the adoption of IBS techniques in Malaysia?

- RQ 2.2: How do stakeholders perceive IBS adoption, and what role do government policies and incentives play in shaping these perceptions?

- RQ 2.3: What are the anticipated trends and potential scenarios for the future adoption of IBS techniques in Malaysia?

RO 3: To determine the optimal coverage radius and planning parameters for cellular base stations serving IBS residential areas, and to compare these parameters with those of conventional (non-IBS) residential zones to identify necessary adjustments in network planning due to IBS construction methods.

- RQ 3.1: How does the optimal coverage radius for cellular base stations vary between IBS and conventional residential areas?

- RQ 3.2: What network planning considerations are required to ensure adequate signal coverage and penetration in IBS-dominated residential zones?

RO 4: To develop policy recommendations and planning guidelines for stakeholders to address indoor connectivity challenges in future IBS developments.

- RQ 4.1: What regulatory or planning interventions can support better indoor connectivity in IBS-built environments?

- RQ 4.2: How can collaboration between developers, telcos, and regulators be structured to proactively address this issue?

(The researcher is invited to propose refinements to Research Objectives and Research Questions. However, the researcher is requested to provide an explanation of the anticipated value of the proposed changes and potential benefits of proposed refinements.)

DI-2 – Assessing Langkawi's Digital Infrastructure Readiness for Inclusive Health, Assisted Living, and Wellness Tourism

A. Research Field:

- Malaysian Communications and Multimedia Commission (MCMC) facilitation of equitable digital infrastructure and communications services delivery.

B. Research Focus:

- MCMC Facilitation and Support of digital health adoption.

C. Research Sponsor:

- Perlis State Office, Northern Regional Office, MCMC.

D. Contact Person for Queries:

- Mr. Mohd Farid Mohamad Heder, Deputy Director, Kedah State Office, Northern Regional Office, farid.mdheder@mcmc.gov.my

E. Research Category:

- Digital Inclusion (DI).
  This study aims to generate evidence to contribute to the development of programme evaluation, assessment and impact of efforts improve access to health, assisted living, wellbeing and health tourism services.

F. Target Groups

1. Stakeholders Benefiting from the Research

The findings and recommendations derived from this research are anticipated to yield benefits for these stakeholder groups:

i. **Malaysian Communications and Multimedia Commission** to obtain insights to guide inclusive infrastructure planning, policy development and regulatory decisions on health-related digital services.

ii. **Health and Wellness Tourism Providers** obtain input for improved digital service delivery, interoperability, and consumer trust (resorts, private hospitals, clinics, wellness centres).

iii. **Local and Foreign Health Seekers** obtain input to improve trust, informed decision-making and improved user experiences via standardised and clear digital information for domestic and international health visitors.

iv. **Government Agencies** obtain robust, evidence-based insights and frameworks supporting digital health tourism growth and policy formulation (Tourism Malaysia, Ministry of Health, Langkawi Development Authority, State Economic Planning Units, Malaysia Healthcare Travel Council).

2. Groups/Communities as Interviewees or Survey Respondents for the Research:

To ensure a comprehensive and well-rounded perspective, the research will engage with a diverse range of stakeholders, who will constitute the primary interviewees or survey respondents:

i. **Local businesses** (wellness and healthcare centres and providers).

ii. **Healthcare professionals** (doctors, nurses, allied health providers).

iii. **Local residents and tourists** (both domestic and international) seeking health and wellness services.

iv. **Telecommunication providers** and local ICT infrastructure developers.

v. **Local government bodies** and **tourism authorities** representatives.

(Where appropriate, researchers are encouraged to propose refinements to respondent categories to better align with the research aims.)

RESEARCH PROBLEM/CONTEXT

Langkawi is progressively shaping its vision to deliver inclusive digital health and assisted living services for its residents, while enhancing its global appeal as a leading destination for wellness and health tourism. This evolving ambition presents a timely opportunity to assess the preparedness and resilience of its digital ecosystem in supporting these dual objectives.

While national initiatives such as JENDELA have laid important groundwork in connectivity, there remains a need for targeted evaluation to determine whether Langkawi's digital infrastructure—encompassing network reliability, service

provider readiness, and user adoption—is sufficiently equipped to meet the specific demands of digital health and wellness services.

This research aligns with the Malaysian Communications and Multimedia Commission's (MCMC) mission to foster a communications and multimedia ecosystem that is inclusive, innovative, and responsive to societal needs. It also supports the broader goals of the National Policy Objectives (NPOs), including equitable access to digital services, infrastructure enhancement, local content and service innovation, and informed participation in the digital economy.

By examining both structural and behavioural dimensions of digital readiness, the study aims to generate actionable insights that can inform infrastructure planning, policy development, and strategic interventions—ultimately contributing to Malaysia's digital transformation and socio-economic progress.

RESEARCH AIMS

The research aims to assess the readiness and quality of Langkawi's digital infrastructure in supporting health and wellness services. It seeks to identify service delivery gaps and adoption barriers experienced by healthcare providers, residents, tourists, and wellness operators. Additionally, the study will evaluate the facilitative role of the MCMC and the effectiveness of existing policy frameworks in driving digital health transformation. Ultimately, the research intends to develop strategic recommendations and propose a comprehensive framework for inclusive digital health, assisted living, wellbeing, and health tourism in Langkawi.

RESEARCH OBJECTIVES

Researchers are invited to submit proposals guided by the following overarching Research Objectives (ROs) and Research Questions (RQs):

RO 1: To evaluate current digital infrastructure readiness in Langkawi for health, assisted living, and wellness services, specifically broadband, mobile connectivity and fixed-line networks.

- RQ 1.1: What is the current state of broadband, mobile, and fixed-line connectivity in Langkawi serving residents, assisted living facilities, and wellness tourism hubs?

- RQ 1.2: Are existing digital platforms and systems interoperable and scalable to meet the needs of healthcare providers, assisted living services, and health and wellness tourists?

RO 2: To access digital literacy and adoption rates among key user groups (Focus: health professionals, wellness operators, assisted living providers, residents, and tourists).

- RQ 2.1: What is the level of digital literacy among health and wellness service providers, including those offering assisted living services?
- RQ 2.2: What are the perceived barriers to digital adoption among local businesses, assisted living operators, and wellness tourism providers?
- RQ 2.3: How do local residents and health/wellness tourists perceive and engage with digital health and wellness services?

RO 3: To identify gaps and barriers to the delivery and adoption of digital health and wellness services (Focus: trust, usability, accessibility, and inclusivity).

- RO 3.1: What are the key barriers to digital service adoption for residents, assisted living recipients, and health tourists?
- RO 3.2: What factors influence trust, usability, and uptake of digital health and wellness services across different user groups?

RO 4: To review current policy frameworks and institutional coordination supporting digital health transformation in Langkawi.

- RQ 4.1: What policies and regulatory frameworks currently support or hinder digital health, assisted living, and wellness service delivery?
- RQ 4.2: Are there gaps in coordination between federal, state, and local agencies that affect service delivery to residents, assisted living communities, and tourists?

RO 5: To develop strategic recommendations for policymakers and stakeholders, proposing targeted interventions to enhance digital readiness and service inclusivity.

- RQ 5.1: What targeted interventions or investments could improve Langkawi's digital readiness for inclusive health, assisted living, and wellness tourism services?

(The researcher is invited to propose refinements to Research Objectives and Research Questions. However, the researcher is requested to provide an explanation of the anticipated value of the proposed changes and potential benefits of proposed refinements.)

DI-3 – Digital Skills Training at NADI: Assessing AI Tutors and Human Educators for Future Workforce Readiness

A. Research Field:

- Digital inclusion and addressing of the digital divide.

B. Research Focus:

- The focus of the research is on using and integrating AI technologies in community-focused digital education.

C. Research Sponsor:

- Perlis State Office, Northern Regional Office, MCMC.

D. Contact Person for Queries:

- Ms. Norrazlina Binti Nordin, Head, Perlis State Office, Northern Regional Office, norrazlina.nordin@mcmc.gov.my

E. Research Category:

- Digital Inclusion (DI).
  This study aims to generate evidence to address the digital divide and to ensure access to quality digital education and skills.

F. Target Groups

1. **Stakeholders Benefiting from the Research**
   The findings and recommendations derived from this research are anticipated to yield benefits for these stakeholder groups:

   i. **NADI Subscribers and Programme Participants** comprising youth and adults engaged in digital skills training, particularly from underserved and rural communities.

   ii. **NADI Trainers and Facilitators** comprising individuals delivering digital literacy and upskilling programmes at NADI centres.

   iii. **Community Members** comprising populations targeted by national digital inclusion efforts, including those in remote or digitally transitioning areas.

iv. **Educators and Curriculum Designers** comprising professionals involved in developing content and pedagogy for community-based digital learning.

v. **Malaysian Communications and Multimedia Commission**, the national regulator overseeing digital inclusion initiatives and NADI programme implementation.

2. Groups/Communities as Interviewees or Respondents for the Research

To ensure a comprehensive and well-rounded perspective, the research will engage with a diverse range of stakeholders, who will constitute the primary interviewees or survey respondents:

i. **Learners and Facilitators**
- **NADI Learners**: Youth and adult participants currently or previously enrolled in digital skills programmes.
- **NADI Trainers and Digital Literacy Facilitators**: On-the-ground educators and support staff involved in programme delivery.

ii. **Technology and Content Contributors**
- **AI Platform Developers and Content Providers**: Technology partners involved in designing or deploying AI-based learning tools.

iii. **Community and Workforce Stakeholders**
- **Community Leaders**: Local influencers and representatives who can speak to community needs, trust, and engagement.
- **Workforce Stakeholders**: Employers, development officers, and industry representatives with insights into future skill requirements.

iv. **Programme Stakeholders**
- **NADI Administrators**: Centre managers and coordinators responsible for day-to-day operations and programme implementation.
- **Deployment Partners**: Organisations contracted to establish and manage NADI centres in collaboration with Internet Service Providers (ISPs).
- **Training and Technology Providers**: Institutions or companies delivering content, certification, or technical support.
- **Collaborative Partners**: Government agencies, NGOs, and private sector entities contributing to digital inclusion, workforce development, or AI integration.

(Where appropriate, researchers are encouraged to propose refinements to respondent categories to better align with the research aims.)

RESEARCH PROBLEM/CONTEXT

As Malaysia advances towards a digitally driven economy, the need for inclusive and effective digital skills training has become increasingly critical, particularly within underserved and digitally transitioning communities. The Pusat Sebaran Maklumat Nasional (NADI) initiative, implemented by the Malaysian Communications and Multimedia Commission (MCMC) under the Universal Service Provision (USP) framework, plays a central role in addressing this need.

NADI centres are designed to provide collective internet access and empower communities—especially in rural and low-coverage areas—through sustainable and relevant programmes. These programmes focus on four key areas: entrepreneurship, lifelong e-learning, digital wellbeing, and maturity in internet use. NADI also serves as a platform for disseminating accurate and timely information on government initiatives such as PADU, Manfaat MADANI, and BUDI MADANI, ensuring that no segment of society is left behind in national development[6].

Despite their growing role, it remains unclear whether AI-powered tutors or traditional human-led instruction are more effective in preparing individuals for the future of work. While AI offers clear advantages in terms of efficiency, scalability, and personalised learning, concerns persist regarding its ability to replicate the emotional intelligence, adaptability, and relational trust that human educators bring to the learning experience.

This study seeks to address a significant evidence gap by assessing both instructional approaches through the lens of learning outcomes, emotional engagement, and stakeholder perceptions. The research is particularly timely given NADI's recent recognition as a WSIS 2025 Champion[7] under the ITU's Capacity Building category, which highlights its success in delivering inclusive digital literacy initiatives to remote communities—Pulau Banggi being a notable example of NADI's ability to reach underserved populations and foster meaningful digital engagement.

Building on this achievement, the study will generate insights to strengthen NADI's delivery model by informing the design of hybrid learning frameworks, supporting scalable deployment of AI-assisted instruction, and guiding infrastructure and curriculum development. These findings will contribute to the enhancement of community-based digital inclusion efforts and reinforce NADI's role in supporting Malaysia's broader digital transformation agenda, by advancing the principles

[6] Malaysian Communications and Multimedia Commission. (2024). Pusat Sebaran Maklumat Nasional (NADI). Accessed on 21 July 2025 from <https://nadi.mcmc.gov.my>

[7] International Telecommunication Union. (2025). WSIS Prizes 2025 – National Information Dissemination Centre (NADI). Accessed on 21 July 2025 at < https://www.itu.int/net4/wsis/stocktaking/Prizes/Prizes/Details/17390840477809796

outlined in the Malaysia Digital Economy Blueprint (MyDIGITAL)[8] and the National 4IR Policy[9], both of which emphasise inclusive digital upskilling, AI adoption, and future workforce readiness across all segments of society.

RESEARCH AIMS

This research aims to assess the comparative effectiveness of AI tutors and human educators in delivering digital skills training at NADI centres. It will examine how each instructional approach influences learning outcomes, emotional engagement, and community perceptions of digital education. The study seeks to determine which model—or combination of models—more effectively supports the development of future-ready digital competencies, particularly within underserved or digitally transitioning communities.

By exploring the interplay between technological innovation and human instruction, the research will contribute valuable evidence to inform the design of inclusive, scalable, and context-sensitive digital learning strategies. These insights will support improvements in NADI's community-based delivery model and guide policy and programme development aimed at equipping Malaysians for work transformations shaped by AI, automation, and evolving employer expectations.

RESEARCH OBJECTIVES

Researchers are invited to submit proposals guided by the following overarching Research Objectives (ROs) and Research Questions (RQs):

RO 1: To assess the learning outcomes and emotional engagement of participants receiving AI tutor-based instruction, human-led instruction, and blended approaches within NADI digital skills programmes.

- RQ 1.1: What differences exist in skill acquisition between AI-led, human-led, and blended instructional models?
- RQ 1.2: Which model demonstrates the most consistent improvement across digital competencies?
- RQ 1.3: How do learners describe their emotional connection and motivation under each instructional approach?
- RQ 1.4: What factors influence sustained participation and interest in digital skills programme?

[8] Economic Planning Unit. (2021). Malaysia Digital Economy Blueprint (MyDIGITAL). Accessed at 21 July 2025 from <https://nagiz.com/wp-content/uploads/2021/02/MyDigital-Malaysia_Digital_Economy_Blueprint.pdf>
[9] Strategic Change Management Office. (2021). Malaysia's Digital Transformation: Malaysia Digital Economy Blueprint and the National 4IR Policy. Accessed at 21 July 2025 from < https://www.mcmc.gov.my/skmmgovmy/media/General/DOS_2021/Session-1_S2_MyDIGITAL.pdf>

RO 2: To explore stakeholders' perceptions – including learners and trainers – regarding the effectiveness, trustworthiness, and acceptability of AI-driven instruction, and is potential to complement human-led teaching.

- RQ 2.1: How do learners and trainers perceive the credibility and reliability of AI tutors?
- RQ 2.2: How do issues of language, cultural relevance, and digital literacy impact AI adoption?
- RQ 2.3: In what ways do stakeholders view AI as a complement to, rather than a replacement for human educators?

RO 3: To analyse contextual challenges and opportunities in deploying AI tutoring within community-based learning environments, with attention to infrastructure, accessibility, equity, and cultural relevance.

- RQ 3.1: What infrastructural or technical barriers affect the deployment of AI tutors in NADI centres?
- RQ 3.2: How can AI tutoring platforms be adapted to accommodate Malaysia's linguistic, ethnic, and cultural diversity in community-based learning environments?
- RQ 3.3: What opportunities exist to adapt AI tools to local learning contexts?

RO 4: To provide strategic recommendations to MCMC and education stakeholders on integrating AI tools responsibly and effectively into digital inclusion and workforce readiness programmes, with emphasis on complementary roles and blended models.

- RQ 4.1: What delivery models best balance scalability, engagement, and trust?
- RQ 4.2: How can AI tools be integrated into existing NADI programmes without displacing human educators?
- RQ 4.3: What policy or operational changes are needed to support blended learning in community-based settings?

(The researcher is invited to propose refinements to Research Objectives and Research Questions. However, the researcher is requested to provide an explanation of the anticipated value of the proposed changes and potential benefits of proposed refinements.)

DI-4 – An Assessment of the Functionality and Connectivity of Master Antenna Television (MATV) Systems in Affordable High-Rise Residential Buildings in the Central Region

A. Research Field:

- Digital Infrastructure Equity and Regulatory Oversight in Malaysia's Broadcasting Ecosystem.

B. Research Focus:

- Evaluation of Master Antenna Television (MATV) System Effectiveness in Delivering Digital Terrestrial Television Services to Residents of Affordable High-Rise Housing in the Central Region.

C. Research Sponsor:

- Broadcasting Development Department, Emerging Technologies Development Division, MCMC.

D. Contact Person for Queries:

- Ts. Mohd Hazmi Husini, Deputy Director, Broadcasting Development Department, Emerging Technologies Development Division, hazmi.husini@mcmc.gov.my

E. Research Category:

- Digital Inclusion (DI).
  This study aims to generate evidence to address the digital divide by evaluating connectivity and its impact on access to broadcast information.

F. Target Groups

1. Stakeholders Benefiting from the Research

The findings and recommendations derived from this research are anticipated to yield benefits for these stakeholder groups:

i. **Digital Terrestrial Television (DTT) and Direct-to-Home (DTH) Infrastructure Providers** will gain accurate insights into current infrastructure usage and awareness gaps, helping to optimise their service outreach and identify underserved buildings.

ii. **Broadcasters** will benefit from clearer data on the reach and barriers to public access or FTA channels.

iii. **Malaysian Communications and Multimedia Commission** will receive ground input to support policy formulation, infrastructure standards, and digital inclusion campaigns related to MATV.

iv. **Local Councils (PBTs)** can use the findings to assess improvement of the MATV systems, building compliance, and prioritise community needs.

v. **Joint Management Bodies (JMBs)** will gain insights into sustainable solutions for maintaining and upgrading MATV systems.

2. Groups/Communities as Interviewees or Respondents for the Research

To ensure a comprehensive and well-rounded perspective, the research will engage with a diverse range of stakeholders, who will constitute the primary interviewees or survey respondents:

i. **Building owners or managers and JMBs** of affordable high-rise residential buildings within the **Central Region**, specifically:
- Selangor
- Wilayah Persekutuan Kuala Lumpur
- Wilayah Persekutuan Putrajaya

ii. **Targeted building types** refer to those constructed from 2017 onward and include, but are not limited to:
- Flats
- Apartments
- Multi-storey *Program Perumahan Rakyat* (PPR)
- *Rumah Selangorku*
- PRIMA affordable housing initiative
- *Perumahan Penjawat Awam* (PPAM)

These stakeholders are directly responsible for infrastructure maintenance and serve as key informants regarding MATV functionality, resident engagement, and service issues. For the purpose of this study, affordable housing is generally defined by national benchmarks set by the **Ministry of Local Government Development (KPKT)** and the **National Property Information Centre (NAPIC)**, which include properties priced at **RM300,000 or below** and serving the **B40 income group** (monthly household income between **RM3,000 and RM4,850**).

(Where appropriate, researchers are encouraged to propose refinements to respondent categories to better align with the research aims.)

RESEARCH PROBLEM/CONTEXT

Free-to-air (FTA) TV services have been made available to the Rakyat via the Digital Terrestrial Television (DTT) platform nationwide. FTA TV reaches large segments of the population, making it a powerful tool for disseminating public information, promoting educational content, and bridging demographic gaps. By providing equal access to information, FTA TV plays a vital role in shaping society, education, and public opinion, especially among underserved communities.

To support the nationwide Analog Switch-Off (ASO) project in 2019, the Malaysian Communications and Multimedia Commission (MCMC) through the Malaysian Technical Standards Forum Bhd (MTFSB) has developed the Technical Code MCMC MTSFB TC G008 – Technical Standard and Infrastructure Requirements for Broadcast Network Facility in 2016. This code was subsequently gazetted in 2017 under the One Stop Centre (OSC) guidelines of the Ministry of Local Government Development (KPKT) and is enforced by local authorities (PBTs) for all new development projects, particularly high-rise buildings, to ensure compliance with communication infrastructure requirements.

The provision of broadcasting infrastructure in high-rise buildings through the Master Antenna Television (MATV) system enables residents to access FTA TV services without incurring additional costs for private antenna purchase or installation. However, constraints in connectivity persist—particularly in the affordable housing category—due to MATV systems that may be malfunctioning, outdated, poorly maintained, or underutilised. There is currently limited public data available on the operational status, user awareness, and integration of MATV systems in these buildings.

In May 2024, MCMC conducted a workshop with selected local councils (PBTs) in Selangor and Kuala Lumpur to gain a deeper understanding of issues related to MATV systems. Findings revealed that most local councils and Joint Management Bodies (JMBs) do not maintain data on the functionality or connectivity of MATV systems within their residential buildings. These systems are often poorly maintained and lack proper monitoring—especially when compared to other shared infrastructure services such as electricity, water, and internet connectivity.

Further inspections by MCMC in selected PPR buildings under the purview of KPKT and DBKL confirmed that many MATV systems were non-functional and had not been properly maintained. Additionally, the absence of structured feedback mechanisms from building owners and JMBs means that issues related to connectivity, service quality, and user satisfaction often go undocumented. This leads to underutilised MATV infrastructure and ineffective delivery of public information.

This study, falling under the Guided Research Category of Digital Inclusion (DI), aims to assess the functionality and connectivity of MATV systems in affordable high-rise residential buildings in the Central Region. The goal is to generate evidence that supports equitable access to information and helps bridge the digital divide through improved infrastructure planning and policy alignment.

RESEARCH AIMS

To assess the functionality, connectivity, and user engagement of Master Antenna Television (MATV) systems in affordable high-rise residential buildings, and to generate evidence-based recommendations for infrastructure improvement, policy development, and digital inclusion. The study also aims to support broadcasting industry sustainability and validate assumptions regarding limited FTA access among low- to middle-income urban populations.

RESEARCH OBJECTIVES

Researchers are invited to submit proposals guided by the following overarching Research Objectives (ROs) and Research Questions (RQs):

RO 1: To assess the operational status, connectivity, and maintenance practices of MATV systems in affordable high-rise residential buildings.

- RQ 1.1: What is the current functional status of MATV systems across selected buildings?
- RQ 1.2: How consistent and reliable is the connectivity provided by these systems?
- RQ 1.3: What maintenance routines are in place for MATV systems?
- RQ 1.4: Are service logs and technical personnel records consistently maintained and accessible?
- RQ 1.5: How do these practices affect system uptime and reliability?

RO 2: To investigate service disruptions, complaint patterns, and resident awareness and usage of MATV systems.

- RQ 2.1: What types of complaints related to MATV systems are most frequently reported?
- RQ 2.2: How are these complaints tracked, addressed, and resolved?

- RQ 2.3: How aware are residents of the availability and benefits of MATV systems?

- RQ 2.4: What proportion of residents actively use MATV for FTA TV and radio access?

- RQ 2.5: What factors influence user engagement or disengagement?

RO 3: To identify financial mechanisms and propose strategies to improve MATV system performance and digital inclusion.

- RQ 3.1: What financial mechanisms, including sinking funds, service fees, and potential government support, are currently used by JMBs and local councils to maintain or upgrade MATV systems?

- RQ 3.2: How are existing financial procedures, such as the use of sinking funds or maintenance fees, applied to address MATV system issues in affordable high-rise buildings?

- RQ 3.3: What strategies can enhance resident awareness and adoption of MATV services?

- RQ 3.4: What infrastructure or policy improvements could support better MATV performance and sustainability?

(The researcher is invited to propose refinements to Research Objectives and Research Questions. However, the researcher is requested to provide an explanation of the anticipated value of the proposed changes and potential benefits of proposed refinements.)

DI-5 – Aligning Connectivity and Development: Evaluating JENDELA's Impact in Rural Sabah Communities with Infrastructure Gaps.

A. Research Field:

- Connectivity, Digital Inclusion and ICT4D (Information and Communication Technologies for Development).

B. Research Focus:

- Assessment of development outcomes and understanding of community outcomes.

C. Research Sponsor:

- Celcom Timur (Sabah) Sdn. Bhd.

D. Contact Person for Queries:

- Mr. Sheridan Hans Peter, Head of Operations and Maintenance, Operations & Maintenance Department, Network Division, Celcom Timur (Sabah) Sdn. Bhd., sheridan@ctsabah.com.my

E. Research Category:

- Guided Research Category of Digital Inclusion.
  This research aims to generate evidence on how JENDELA-enabled connectivity in rural Sabah influences socio-economic impact amid ongoing infrastructure gaps.

F. Target Groups

1. **Stakeholders Benefiting from the Research**
   The findings and recommendations derived from this research are anticipated to yield benefits for these stakeholder groups:

   i. **Telecommunications Companies and Infrastructure Partners in Sabah** to inform service delivery, tower deployment, and rural connectivity strategies.

   ii. **Malaysian Communications and Multimedia Commission (MCMC)** to support policy refinement, monitoring frameworks, and digital inclusion strategies.

iii. **State Development Agencies and Policymakers** to support integrated planning that aligns digital infrastructure with broader socio-economic development priorities.

2. Groups/Communities as Interviewees or Respondents for the Research

To ensure a comprehensive and well-rounded perspective, the research will engage with a diverse range of stakeholders, who will constitute the primary interviewees or survey respondents:

i. **Residents in rural Sabah communities**
   - Individuals living in areas connected under JENDELA but underserved in terms of basic infrastructure (e.g., roads, water, electricity).
   - Focus on diverse demographics including youth, elders, small business owners, and students.

ii. **Local leaders and civil society groups**
   - Village heads, community organisers, and NGOs involved in digital literacy, rural development, or advocacy.
   - Key informants for understanding community-level mobilisation and perceptions.

iii. **Community engagement teams from telecommunications companies in Sabah, MCMC, and state development agencies**
   - Personnel involved in outreach, implementation, and monitoring of connectivity projects.
   - Insights into planning, challenges, and alignment with development goals.

(Where appropriate, researchers are encouraged to propose refinements to respondent categories to better align with the research aims.)

RESEARCH PROBLEM/CONTEXT

Malaysia's JENDELA initiative has considerably expanded digital connectivity across rural areas of Sabah, including communities that continue to lack essential infrastructure such as surfaced roads, piped water, and reliable electricity. This uneven development landscape presents a planning challenge: while internet access is now available in many remote locations, the absence of foundational services may influence how effectively digital inclusion translates into tangible socio-economic benefits.

This research is situated within the evolving context of "connectivity-first" rollouts, where digital infrastructure is introduced ahead of other basic services. Such an

approach presents both opportunities and limitations. On one hand, connectivity can serve as a catalyst for access to education, healthcare, economic participation, and civic engagement. On the other, its impact may be constrained when supporting infrastructure and services are not yet in place.

Recent federal efforts have acknowledged these development gaps. In May 2024, the government allocated RM1.14 billion to Sabah for rural infrastructure development, with RM1.06 billion designated for 224 projects involving roads, bridges, water, and electricity supply (Miwil, 2024) [10]. However, despite substantial allocations, progress on many infrastructure projects has been slow, highlighting the ongoing challenges in translating budgetary commitments into improvements on the ground.

This study will assess both the socio-economic outcomes and community perceptions of JENDELA-enabled connectivity in rural Sabah. It will explore whether internet access has facilitated improvements in public services and livelihoods, and whether residents perceive these changes as meaningful or limited by ongoing infrastructure deficits. It will also examine how digital platforms are being used to amplify local voices—through expressions of concern, community mobilisation, or engagement with public institutions.

Ultimately, the research aims to support Sabah-based telecommunication companies, MCMC, and other development stakeholders by providing insights into how connectivity-first approaches are experienced in practice. The findings will help guide future rollout prioritisation, community engagement, and inter-agency planning by identifying where fibre connectivity can catalyse impact—and where complementary infrastructure and support may be needed to maximise outcomes.

RESEARCH AIMS

To examine how digital connectivity introduced through the JENDELA initiative is influencing socio-economic outcomes and community perceptions in rural Sabah communities that continue to face significant development and infrastructure gaps. This study will explore the extent to which internet access contributes to improvements in education, healthcare, government services, and economic participation in areas where basic infrastructure—such as roads, piped water, and reliable electricity—remains limited. The research aims to provide practical insights for Celcom Timur (Sabah) Sdn Bhd, Sabah-based telcos, and key federal and state stakeholders by identifying how connectivity is being used, where its impact is constrained, and how future infrastructure planning can better align digital inclusion with broader development priorities.

[10] Miwil, O. (2024, May 9). Sabah allocated RM1.14 billion for development of rural infrastructure, says Zahid. New Straits Times. Accessed on 28 July 2025 at <https://www.nst.com.my/news/nation/2024/05/1048780/sabah-allocated-rm114-billion-development-rural-infrastructure-says>

RESEARCH OBJECTIVES

Researchers are invited to submit proposals guided by the following overarching Research Objectives (ROs) and Research Questions (RQs):

RO 1: To identify and profile rural communities in Sabah with fibre or mobile connectivity but lacking access to basic infrastructure such as roads, piped water, or stable electricity.

- RQ 1.1: Which rural communities in Sabah have received digital connectivity under JENDELA?
- RQ 1.2: What types of basic infrastructure are still lacking in these connected communities?

RO 2: To evaluate whether digital connectivity has contributed to improved access to government services, education, economic activity, or healthcare.

- RQ 2.1: Has internet access facilitated better access to government services in these communities?
- RQ 2.2: What impact has connectivity had on education, healthcare, and local economic activity?

RO 3: To assess residents' perceptions of digital inclusion, including perceived benefits and frustrations related to unmet infrastructure needs.

- RQ 3.1: How do residents perceive the benefits of digital connectivity in their daily lives?
- RQ 3.2: Do residents express frustration or limitations due to the absence of basic infrastructure?

RO 4: To examine how digital platforms are being used to raise local issues, mobilise communities, or engage with public institutions.

- RQ 4.1: Are residents using digital platforms to voice concerns or complaints?
- RQ 4.2: How are digital tools being used for community mobilisation or engagement with authorities?

RO 5: To recommend strategies for Celcom Timur (Sabah) Sdn. Bhd., MCMC, and state agencies to better align digital connectivity efforts with broader development priorities.

- RQ 5.1: What gaps exist between current connectivity efforts and broader development needs?

- RQ 5.2: What strategies could improve coordination between digital infrastructure planning and socio-economic development?

(The researcher is invited to propose refinements to Research Objectives and Research Questions. However, the researcher is requested to provide an explanation of the anticipated value of the proposed changes and potential benefits of proposed refinements.)

DI-6 – Positioning 5G as a Public Utility in Sabah: Strategies for Inclusive and Impactful Connectivity

A. Research Field:

- Digital Inclusion: Ensuring Connectivity and Addressing the Digital Divide.

B. Research Focus:

- 5G adoption strategies and measures.

C. Research Sponsor:

- Celcom Timur (Sabah) Sdn. Bhd.

D. Contact Person for Queries:

- Mr. Loy Cheah Cheong, Head of Planning & Design, Network Planning & Design Department, Network Division, Celcom Timur (Sabah) Sdn. Bhd., loy@ctsabah.com.my

E. Research Category:

- Guided Research Category of Digital Inclusion.
  This research aims to generate evidence to support equitable connectivity by identifying strategies and measures for 5G adoption in Sabah. The study focuses on addressing the digital divide, particularly among underserved communities, in ways that enable meaningful participation in digitalisation and society.

F. Target Groups

1. **Stakeholders Benefiting from the Research**
   The findings and recommendations derived from this research are anticipated to yield benefits for these stakeholder groups:

   i. **Local Communities and Leaders (Semi-urban and Rural Sabah)** will benefit from improved service reliability and community empowerment through awareness and engagement in digital inclusion and infrastructure planning.

   ii. **B40 Youth and Gig Workers in Sabah** will gain access to more affordable and relevant 5G-enabled services, supporting digital livelihoods and social mobility.

iii. **Micro-SMEs and Small Business Owners in Sabah** will benefit from insights into how 5G can support business analytics, digital marketing, and operational efficiency.

iv. **Education and Health Sector Users in Sabah** will be supported through improved access to remote learning, telehealth, and smart service delivery powered by 5G infrastructure.

v. **Sabah Telecommunications Companies** will benefit from clearer policy frameworks and strategic recommendations to support 5G mainstreaming as a public utility.

vi. **Government Agencies and Regulatory Bodies in Sabah (e.g., MCMC, State Ministries, Local Authorities)** will benefit from clearer legal frameworks and enforcement guidelines, enhancing their capacity to respond effectively to infrastructure-related offences.

2. Groups/Communities as Interviewees or Respondents for the Research

To ensure a comprehensive and well-rounded perspective, the research will engage with a diverse range of stakeholders, who will constitute the primary interviewees or survey respondents:

i. **Local Community Members and Leaders (Semi-Urban and Rural Sabah)**
Can share lived experiences and perceptions, contributing to a better understanding of community-level challenges and attitudes toward 5G.

ii. **B40 Youth and Gig Workers in Sabah and Local Community Members and Leaders**
Can provide insights into affordability, relevance, and barriers to adoption of 5G services in their daily lives.

iii. **Micro-SME Owners and Operators in Sabah**
Can reflect on the potential and limitations of 5G in supporting business operations, digital tools, and market access.

iv. **Students, Teachers, and Healthcare Practitioners in Sabah**
Can provide sector-specific perspectives on how 5G impacts service delivery, learning, and public health outcomes.

v. **Telecommunications Users in Sabah**
Can offer feedback on service reliability, infrastructure gaps, and regional disparities in access.

vi. **Sabah Telecommunications Companies Network Operations and Legal Teams**
Can contribute operational, technical, and strategic insights into 5G deployment, service design, and regulatory challenges.

vii. **Government and Regulatory Officials in Sabah (e.g., MCMC, Local Authorities)**
Can reflect on policy implementation, enforcement challenges, and opportunities for regulatory innovation.

(Where appropriate, researchers are encouraged to propose refinements to respondent categories to better align with the research aims.)

RESEARCH PROBLEM/CONTEXT

Sabah's participation in Malaysia's national 5G rollout marks a significant step toward digital transformation. However, public adoption in the state is still largely shaped by the prevailing view that 5G is primarily about faster smartphone connectivity. This limited perception may constrain the technology's perceived value among key societal segments, including the general public in semi-urban and rural areas, B40 youth and gig workers, micro-SMEs, and users in education and health sectors. In certain regions of Sabah, telco customers may also face challenges related to infrastructure equity and service reliability, influenced by geographic and provider-specific factors.

Unlike West Malaysia, where the Uniform Building By-Laws (UBBL 2022) formally recognise communications infrastructure as a public utility (Government of Malaysia, 2022)[11], Sabah operates under its own legal framework. Governed by the Sabah Land Ordinance (Cap 68), land use in the state has recently been amended to allow reservation for public purposes, including communications infrastructure (Government of Sabah, 2025)[12] . These developments signal a growing policy alignment with core public utility principles such as essentiality, universality, reliability, and public benefit.

[11] Government of Malaysia. (2021). Uniform Building By-Laws (Amendment 2021). Jabatan Kerajaan Tempatan, Kementerian Pembangunan Kerajaan Tempatan. Accessed on 31 July at <https://jkt.kpkt.gov.my/pindaan-undang-undang-kecil-bangunan-seragam-1984-ukbs-1984-yang-di-warta-mengikut-pihak-berkuasa/>

[12] Government of Sabah. (2025). Warta Kerajaan Sabah: Gazette Notification No. 003133. Accessed on 31 July at <https://sabah.gov.my/gazette/docs/003133.pdf>

Nonetheless, the practical rollout of 5G in Sabah may not yet fully embody these principles. Usage dimensions — including affordability, relevance to daily life, digital literacy, and localised content — remain under-addressed, particularly for underserved communities. This research seeks to bridge the gap between policy intent and lived reality by exploring how 5G can be operationalised as a true public utility in Sabah — accessible, relevant, and impactful across diverse societal segments.

RESEARCH AIMS

To investigate the socio-technical barriers and enablers of mainstream 5G adoption in Malaysia, and to develop strategic recommendations for positioning 5G as a widely accessible and socially valuable utility.

RESEARCH OBJECTIVES

Researchers are invited to submit proposals guided by the following overarching research objectives (ROs) and Research Questions (RQs):

RO 1: To examine public perceptions, awareness, and usage patterns of 5G across diverse demographic and geographic segments.

- RQ 1.1: What are the prevailing public perceptions of 5G in Malaysia?
- RQ 1.2: How do usage patterns of 5G vary across different regions, age groups, income levels, and education backgrounds?
- RQ 1.3: What factors influence public understanding and engagement with 5G beyond perceived internet performance (e.g., speed, latency) and data limitations (e.g. caps, quotas)?

RO 2: To evaluate the accessibility and relevance of existing 5G-powered services in sectors such as education, healthcare, agriculture, public safety, and microenterprise.

- RQ 2.1 – What 5G-enabled services are currently available in Malaysia across key sectors?
- RQ 2.2 – How accessible are these services to different societal groups, including rural communities and small businesses?

- RQ 2.3 – To what extent do these services address local needs and contribute to social value?

RO 3: To identify key barriers to 5G adoption, including cost, device availability, digital literacy, and service relevance.

- RQ 3.1: What are the main socio-economic and infrastructural barriers to 5G adoption in Malaysia?
- RQ 3.2: How do issues such as affordability, device compatibility, and digital literacy affect uptake?
- RQ 3.3: What role does content relevance and service availability play in limiting broader 5G use?

RO 4: To analyse international case studies where 5G has been successfully mainstreamed for public benefit.

- RQ 4.1: Which countries have effectively positioned 5G as a public utility, and what strategies did they use?
- RQ 4.2: What lessons can Malaysia learn from these international experiences in terms of policy, outreach, and innovation?

RO 5: To propose policy, communication, and partnership strategies to reposition 5G as a national utility with inclusive societal impact.

- RQ 5.1: To propose policy, communication, and partnership strategies to reposition 5G as a national utility with inclusive societal impact.
- RQ 5.2: How can communication strategies be tailored to promote 5G as a socially valuable utility?
- RQ 5.3: What types of partnerships (e.g., public-private, community-based) are needed to drive inclusive 5G adoption?

(The researcher is invited to propose refinements to Research Objectives and Research Questions. However, the researcher is requested to provide an explanation of the anticipated value of the proposed changes and potential benefits of proposed refinements.)

DI-7 – Enabling FTTH Adoption in Brownfield Zones: A Sabah Case Study on Infrastructure Diversity and Consumer Choice

A. Research Field:

- Digital Inclusion: Ensuring Connectivity and Addressing the Digital Divide.

B. Research Focus:

- Fibre-To-The-Home (FTTH) adoption strategies and measures.

C. Research Sponsor:

- Celcom Timur (Sabah) Sdn. Bhd.

D. Contact Person for Queries:

- Mr. Mahmood Umar, Head of Fixed Broadband, Business Development Division, Celcom Timur (Sabah) Sdn. Bhd., mahmoodumar@ctsabah.com.my

E. Research Category:

- Guided Research Category of Digital Inclusion.
  This research aims to generate evidence to support digital inclusion by identifying strategies and measures for FTTH adoption in Sabah. The study focuses on factors for ensuring equitable connectivity and addressing the digital divide and provide insights to ensure the sustainability on non-incumbent infrastructure providers.

F. Target Groups

1. **Stakeholders Benefiting from the Research:**
   The findings and recommendations derived from this research are anticipated to yield benefits for these stakeholder groups:

   i. **Government and Regulatory Bodies**
   - **MCMC and relevant government agencies** involved in digital infrastructure planning and policy.
   - **Policymakers and regulators** aiming to promote fair competition, infrastructure diversity, and digital inclusion under the CMA.

ii. **Industry Stakeholders in Sabah**
- **Non-incumbent infrastructure providers** seeking to improve FTTH take-up in brownfield zones.
- **ISPs** exploring alternative infrastructure partnerships and commercial models in competitive markets.

3. Groups/Communities as Interviewees or Respondents for the Research

To ensure a comprehensive and well-rounded perspective, the research will engage with a diverse range of stakeholders, who will constitute the primary interviewees or survey respondents:

i. **End-Users**
- **Residential and business users** in brownfield areas where FTTH infrastructure is available.
- **Consumers** who either subscribed to the incumbent provider or chose not to subscribe to any FTTH service.

ii. **Industry and Community Stakeholders**
- **ISPs** operating in Sabah that partner with infrastructure providers.
- **Property developers or building managers** with experience accommodating multiple FTTH providers.
- **Community members and representatives** in fibre-ready areas with limited service options or low awareness.

(Where appropriate, researchers are encouraged to propose refinements to respondent categories to better align with the research aims.)

RESEARCH PROBLEM/CONTEXT

In Sabah's brownfield areas—where Fibre-To-The-Home (FTTH) infrastructure has already been deployed by incumbent providers—non-incumbent providers face significantly lower adoption rates despite comparable network readiness. This limited uptake is shaped by factors such as consumer awareness, brand loyalty, bundled service lock-ins, and pricing perceptions.

While these challenges are not unique to Sabah, the region's distinct regulatory and market environment, coupled with its digital inclusion needs, make it a critical case for understanding how infrastructure diversity can be translated into real consumer choice. The Communications and Multimedia Act (CMA) mandates MCMC to promote fair competition and prevent market dominance. This research is intended to complement

existing policy objectives by exploring factors that may influence FTTH adoption in brownfield zones. It aims to identify insights that may inform inclusive market strategies and infrastructure planning, while remaining sensitive to the complexities of existing market dynamics and provider roles.

RESEARCH AIMS

To investigate the factors contributing to low FTTH adoption in brownfield areas with existing infrastructure, particularly in contexts where the provider is not the incumbent. The study aims to identify both demand-side and supply-side barriers and to develop strategic recommendations that enhance adoption rates for non-incumbent infrastructure providers, while supporting broader goals of digital inclusion and fair competition.

RESEARCH OBJECTIVES

Researchers are invited to submit proposals guided by the following overarching Research Objectives (ROs) and Research Questions (RQs):

RO 1: To identify the key reasons customers in brownfield areas do not subscribe to FTTH services from non-incumbent providers, despite having infrastructure access.

- RQ 1.1: What are the most cited reasons by consumers for not adopting FTTH services from non-incumbent providers?
- RQ 1.2: How do perceptions of service quality and reliability differ between incumbent and non-incumbent providers?
- RQ 1.3: What role does infrastructure visibility and consumer awareness play in adoption decisions?

RO 2: To analyse consumer decision-making factors (pricing, trust, brand familiarity, and service bundling) that influence take-up.

- RQ 2.1: How do pricing models influence consumer choice between incumbent and non-incumbent providers?
- RQ 2.2: To what extent does brand familiarity and trust affect FTTH subscription decisions?
- RQ 2.3: How do bundled services (e.g., broadband + mobile + TV) impact consumer loyalty and switching behavior?

RO 3: To evaluate the effectiveness of current commercial models and engagement strategies used by non-incumbent providers in similar markets.

- RQ 3.1: What commercial models are currently employed by non-incumbent providers in brownfield zones?
- RQ 3.2: How effective are these models in driving adoption compared to incumbent strategies?
- RQ 3.3: What community engagement or outreach efforts have shown measurable impact in similar contexts?

RO 4: To recommend targeted actions that second providers can take—including pricing strategies, ISP collaboration models, and community engagement—to improve FTTH adoption in brownfield zones.

- RQ 4.1: What pricing strategies are most effective in attracting users from incumbent providers?
- RQ 4.2: How can collaboration with ISPs or local stakeholders enhance market penetration?
- RQ 4.3: What community engagement approaches are most likely to build trust and drive adoption?

(The researcher is invited to propose refinements to Research Objectives and Research Questions. However, the researcher is requested to provide an explanation of the anticipated value of the proposed changes and potential benefits of proposed refinements.)

FW-1– Building a Sustainable Future of Work: Assessing Flexible Working Hours in the Sabah Telecommunications Sector

A. Research Field:

- Human Resources Management and Organisational Studies.

B. Research Focus:

- Telecommunications sector flexible work adoption challenges and opportunities.

C. Research Sponsor:

- Celcom Timur (Sabah) Sdn. Bhd.

D. Contact Person for Queries:

- Ms. Natasha Assan, Head of Human Resources, Human Resources Department, Human Resources, Administration & Compliance Division, Celcom Timur (Sabah) Sdn. Bhd, natasha@ctsabah.com.my

E. Research Category:

- Future of Work.
  This research aims to generate evidence that will aid in the assessment of the adoption of Flexible Working Arrangements (FWA) within the telecommunications sector.

F. Target Groups

1. **Stakeholders Benefiting from the Research:**
   The findings and recommendations derived from this research are anticipated to yield benefits for these stakeholder groups:

   i. **Telecommunications Companies in Sabah** will gain insights to improve workforce and operational efficiency for telecommunications providers throughout Malaysia.

   ii. **Senior Leadership and Management** will gain insights to support long-term planning with data-driven recommendations for flexible work models.

   iii. **HR and Policy Makers** will gain insights to develop informed policies that enhance employee well-being and regulatory compliance.

iv. **Line Managers and Supervisors** will gain insights on practical tools to manage flexible teams and improve team performance.

v. **Employees** will gain insights related to work-life balance, job satisfaction, and mental well-being.

vi. **Government, Regulators and Industry Associations** will obtain empirical evidence to guide sector-wide standards and policy development.

2. Groups/Communities as Interviewees or Survey Respondents for the Research

To ensure a comprehensive and well-rounded perspective, the research will engage with a diverse range of stakeholders, who will constitute the primary interviewees or survey respondents:

i. **Organisational Decision Makers**
- Understand strategic priorities, policy readiness, leadership alignment, and governance expectations.
  - Senior Leadership and Management
  - HR and Policy Makers
  - Line Managers and Supervisors
  - Regulators and Industry Associations

ii. **Operational Workforce**
- Assess cultural readiness, job role suitability, and employee expectations for FWA.
  - Operational Employees
  - Support Function Employees
  - Employee Representatives

iii. **Enabling Functions**
- Identify systems, controls and infrastructure needed to support FWA implementation
  - Human Resources
  - IT teams
  - Compliance teams

iv. **Institutional Stakeholders**
- Evaluate sector-wide impact, policy alignment, and scalability of FWA models.
  - Telecommunications Companies in Sabah
  - Government, Regulators and Industry Associations

(Where appropriate, researchers are encouraged to propose refinements to respondent categories to better align with the research aims.)

RESEARCH PROBLEM/CONTEXT

Flexible Working Arrangements (FWA) are increasingly recognised as a strategic imperative in the evolving landscape of work. In Sabah's telecommunications sector, however, their adoption remains informal and inconsistent, lacking a standardised policy framework. This gap presents a critical need for research to guide evidence-based implementation.

Despite widespread digital connectivity that makes flexible work technically feasible, organisations face significant challenges in operationalising FWA. These include ensuring accountability, maintaining service standards, and identifying job roles best suited for flexible arrangements. Without clear governance structures, organisations risk inefficiencies, compliance issues, and reputational harm. Moreover, workforce expectations have shifted dramatically post-pandemic. Employees now prioritise autonomy, work-life balance, and flexibility. Organisations that fail to adapt risk losing talent, experiencing lower engagement, and falling behind in competitiveness. Yet, readiness for this transition is uneven—many organisations lack the cultural maturity and leadership alignment needed to support output-based work models.

In Malaysia, the Ministry of Human Resources has reinforced the importance of FWA through amendments to the Employment Act 1955, granting employees the right to formally request flexible arrangements[13]. Recent surveys, such as those by Hays Malaysia, show that employees working fully on-site are less satisfied with their arrangements compared to those with flexible options, highlighting a growing preference for autonomy and hybrid models[14].

Adopting FWA also aligns with broader strategic goals. It supports sustainability and ESG commitments by reducing commuting, lowering carbon footprints, and promoting inclusive work practices. It contributes to national digital transformation agendas and enhances organisational resilience.

Sabah's unique geographic and socio-economic context further underscores the need for tailored solutions. A one-size-fits-all approach may not be effective. Localised research is essential to uncover context-sensitive models that balance operational needs with employee wellbeing. This study will therefore assess the feasibility, cultural readiness, and governance requirements for implementing sustainable flexible working hours in Sabah's telecommunications sector—providing actionable insights to support informed decision-making and strategic transformation.

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[13] Ministry of Huma Resources (KESUMA), *Sections 60 (p) (q) of the Employment Act 1955 (Act 265).* Accessed on 28 July 2025 at <https://jtksm.mohr.gov.my/sites/default/files/2023-11/Akta Kerja 1955 %28Akta 265%29.pdf>

[14] Hays (2024). *Flexible Working in 2024.* Accessed on 28 July 2024 at <https://www.hays.com.my/press-release/content/flex-working-in-2024-1>

RESEARCH AIMS

This study aims to support informed decision-making by assessing the feasibility, cultural readiness, and governance requirements for implementing sustainable flexible working hours in Sabah's telecommunications sector. It will examine operational needs, leadership alignment, and employee expectations to develop context-sensitive frameworks that promote productivity, inclusion, and strategic alignment with ESG and national digital transformation goals.

RESEARCH OBJECTIVES

Researchers are invited to submit proposals guided by the following overarching Research Objectives (ROs) and Research Questions (RQs):

RO 1: To evaluate the feasibility of implementing flexible working hours across job roles in Sabah's telecommunications sector.

- RQ 1.1: Which job roles are most suitable for flexible working hours in Sabah's telecommunications sector?
- RQ 1.2: What are the perceived benefits and challenges of flexible working arrangements from both employer and employee perspectives?
- RQ 1.3: What are the current informal practices or precedents for flexible work in the sector, and how do they inform future feasibility?

RO 2: To identify the operational, policy, and cultural requirements—including leadership alignment—necessary for transitioning to sustainable flexible working hours.

- RQ 2.1: What systems, policies, and accountability mechanisms are needed to support flexible work while maintaining service standards?
- RQ 2.2: How culturally ready are organisations and employees in Sabah's telecommunications sector to shift from attendance-based to output-based work models?
- RQ 2.3: What leadership behaviours, values, and communication practices are essential for enabling and sustaining flexible work environments?

RO 3: To develop a practical framework for FWA governance, including systems, controls, and KPIs tailored to hybrid and remote work.

- RQ 3.1: What governance structures and performance indicators are most effective in managing hybrid and remote work environments?

- RQ 3.2: How can organisations ensure compliance, accountability, and continuous improvement in flexible work models?

- RQ 3.3: How can the proposed framework align with broader ESG goals and organisational sustainability strategies

(The researcher is invited to propose refinements to Research Objectives and Research Questions. However, the researcher is requested to provide an explanation of the anticipated value of the proposed changes and potential benefits of proposed refinements.)

SESG-1 – Catalysing ESG Integration in the Communications and Multimedia Industry: MCMC's Role in Policy Development and Best Practice Facilitation

A. Research Field:

- Malaysian Communications and Multimedia Commission Environmental, Social, and Governance (ESG) adoption.

B. Research Focus:

- MCMC Policy Development and Facilitation of Industry ESG adoption Framework adoption.

C. Research Sponsor:

- Sustainability Intelligence and Research Department, Sustainability Division, MCMC.

D. Contact Person for Queries:

- Ms. Siti Syarizan Mohd Nizom, Deputy Director, Sustainability Intelligence and Research Department, Sustainability Division, syarizan.nizom@mcmc.gov.my

E. Research Category:

- Sustainability & Environment, Social, and Governance (SESG).
  This research aims to generate evidence that will aid in the development of MCMC policies and facilitation of ESG adoption amongst Communications and Multimedia (C&M) industry players.

F. Target Groups

3. Stakeholders Benefiting from the Research

The findings and recommendations derived from this research are anticipated to yield benefits for these stakeholder groups:

i. **Malaysian Communications and Multimedia Commission (MCMC):** As the direct beneficiary, MCMC will gain a robust analytical framework and actionable, evidence-based recommendations to enhance its policy development, regulatory enforcement, and strategic stakeholder engagement in driving ESG integration across the C&M industry. This

will strengthen MCMC's position as a forward-thinking, responsible regulator.

ii. **Communications & Multimedia (C&M) Industry Players**: Regulated entities, including telecommunication service providers, broadcasters, content platforms, and digital infrastructure operators, will benefit from clearer ESG guidelines, standardised best practices, and a more predictable regulatory landscape, fostering sustainable growth and improved compliance.

iii. **Investors and Financial Institutions**: Enhanced ESG clarity and consistent industry-wide integration, catalysed by MCMC, will provide greater transparency and predictability for investors, enabling more informed capital allocation towards sustainable C&M entities.

iv. **Malaysian Government Agencies and Policymakers**: Broader government bodies involved in national sustainability agendas (e.g., Ministry of Natural Resources and Environmental Sustainability, Ministry of Digital, Energy Commission, Securities Commission) may leverage MCMC's refined approach as a model for ESG integration in other regulated sectors.

v. **Consumers and the General Public**: Ultimately, a more sustainably managed and ethically governed C&M industry will lead to benefits such as reduced environmental impact from digital services, enhanced data privacy and ethical technology use, and increased public trust in the digital ecosystem.

4. Groups/Communities as Interviewees or Survey Respondents for the Research

To ensure a comprehensive and well-rounded perspective, the research will engage with a diverse range of stakeholders, who will constitute the primary interviewees or survey respondents:

i. **MCMC Internal Stakeholders:**

- **Regulatory and Policy Divisions**: Representatives from Licensing, Spectrum Management, Compliance, Consumer Affairs, and Universal Service Provision units, possessing insights into current regulatory approaches and challenges.

- **Sustainability Intelligence and Research Department**: Key personnel from the sponsoring department, to leverage their expertise in internal ESG and organisational change.

- **Senior Leadership**: Directors and Divisional Heads, for strategic insights and validation of proposed recommendations.

ii. **Communications & Multimedia (C&M) Industry Representatives:**

- **Sustainability/ESG Leads**: Senior managers or directors responsible for ESG strategy and implementation within major C&M companies in Malaysia.

- **Compliance and Legal Counsel**: Professionals from regulated entities familiar with current regulatory obligations and challenges in ESG reporting and practice.

- **Industry Associations**: Representatives from relevant bodies such as the Communications and Multimedia Consumer Forum of Malaysia (CFM) and industry-specific trade associations, providing a collective industry perspective.

iii. **ESG Experts and Consultants**: Independent specialists with deep knowledge in ESG frameworks, sustainable finance, and regulatory trends relevant to the digital sector.

iv. **Academic Researchers**: Scholars from Malaysian universities specialising in telecommunications policy, environmental studies, social sciences, and corporate governance.

v. **Selected Representatives from Financial and Investor Communities**: Analysts or fund managers with a focus on ESG investments in the C&M industry, to understand investor expectations and market dynamics**.**

(Where appropriate, researchers are encouraged to propose refinements to respondent categories to better align with the research aims.)

RESEARCH PROBLEM/CONTEXT

The global exigency for sustainable development has profoundly amplified the salience of Environmental, Social, and Governance (ESG) considerations across all industrial sectors. Stakeholders, spanning institutional investors, discerning consumers, and authoritative regulatory bodies, are increasingly demanding demonstrable transparency and heightened accountability regarding corporate non-financial performance. Within Malaysia, this escalating imperative is unequivocally underscored by the nation's ambitious commitments, notably articulated in the **National Energy Transition Roadmap (NETR)**, which targets

a 70% renewable energy capacity by 2050[15], and the recently enacted **Energy Efficiency and Conservation Act (EECA)** 2024, mandating energy efficiency across energy-intensive sectors pertinent to digital infrastructure[16]. Furthermore, the Securities Commission Malaysia (SCM) has progressively fortified its ESG reporting mandates for listed entities, epitomised by the formal adoption of the **IFRS S1 and S2 (ISSB standards)** via the **National Sustainability Reporting Guide (NSRF)**[17], thereby exerting substantial influence over major constituents within the Communications & Multimedia (C&M) industry[18].

While prominent Malaysian telecommunication service providers—such as CelcomDigi, Telekom Malaysia (TM), Maxis, and Axiata—have commenced public disclosure of their ESG initiatives and achieved varying ESG ratings, the comprehensive and systematic integration of ESG principles across the entire C&M industry, particularly propelled by a harmonised regulatory policy, remains at a nascent stage. Current industry endeavours, whilst commendable in their individual scope, largely reflect disparate corporate strategies rather than a unified, sector-wide impetus.

The Malaysian Communications and Multimedia Commission (MCMC), as the principal regulatory authority for the C&M industry, is uniquely positioned to act as a pivotal catalyst and guiding force for this critical integration. MCMC has already implemented foundational measures pertaining to specific facets of sustainability, exemplified by the **MCMC MTSFB TC G004:2024 Specification for Green Data Centres**[19], which furnishes essential guidelines for enhancing energy efficiency within digital infrastructure. Concurrently, strategic initiatives such as the JENDELA programme are actively expanding digital connectivity, inherently addressing aspects of social inclusion and, through facilitated infrastructure sharing, yielding environmental co-benefits such as reduced visual pollution and optimised energy consumption[20]. MCMC's proactive stance in data governance, demonstrated by its recent efforts to enable the anonymised use of mobile phone data for statistical purposes, further underscores its evolving role in governing the social and ethical dimensions of digital technologies.

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[15] Ministry of Economy Malaysia. (2023). Summary of the National Energy Transition Roadmap Phase 1. PwC Malaysia. Accessed on 1 July 2025 at <https://www.pwc.com/my/en/assets/publications/2023/pwc-malaysia-summary-national-energy-transition-roadmap-netr-phase-1-august-2023.pdf>

[16] Government of Malaysia. (2024). Energy Efficiency and Conservation Act 2024 (Act 861). Suruhanjaya Tenaga. Accessed on 1 July 2025 at <https://www.st.gov.my/contents/2024/EECA/BI%20-%20Energy%20Efficiency%20and%20Conservation%20Act%202024%20-%20Act%20861.pdf>

[17] Securities Commission Malaysia. (2024). National Sustainability Reporting Framework (NSRF). Accessed on 1 July 2025 at<https://www.sc.com.my/nsrf>

[18] Azmi & Associates. (5 May 2025). "An Overview of ESG in Malaysia in 2025 - Azmi & Associates," Accessed on 1 July 2025 at<https://www.azmilaw.com/insights/an-overview-of-esg-in-malaysia-in-2025/>

[19] Malaysian Technical Standards Forum Berhad (MTFSB) (2024). Technical Code: Specification for Green Data Centres (First Revision) MCMC MTFSB TC G004 2024. Accessed on 1 July 2025 at <MTSFB2307R1_Specifications for Green Data Centre - First Revision>

[20] Malaysian Communications and Multimedia Commission. (n.d.). JENDELA: Jalinan Digital Negara. Accessed on 1 July 2025 at <https://jendela.my>

However, a discernible lacuna exists in the transition from these fragmented, albeit vital, initiatives towards a cohesive and overarching policy framework that comprehensively mandates, incentivises, and strategically facilitates robust ESG integration across the entirety of the C&M value chain. This necessitates a concerted focus on mitigating the multifaceted environmental impacts of network expansion and electronic waste, championing the ethical deployment of advanced technologies (e.g., AI governance, data ethics), and cultivating exemplary corporate governance and stakeholder accountability within its regulated entities.

Without a clear, systematic regulatory directive and robust facilitative mechanisms from MCMC, the industry's ESG progression may remain disparate, thereby potentially impeding Malaysia's overarching national sustainability objectives and risking adverse impacts on national competitiveness and public trust within the digital sector. This proposed research endeavours to critically analyse and identify how MCMC can most effectively discharge this pivotal catalytic role, thereby furnishing actionable policy and best practice recommendations for both the industry and for MCMC's own internal organisational evolution.

RESEARCH AIMS

This research provides MCMC with a robust framework to catalyse comprehensive ESG integration across the Malaysian C&M industry. It will critically assess current ESG integration, evaluating MCMC's existing and potential roles in fostering sustainability performance. The project identifies and analyses effective policy instruments, regulatory frameworks, and best practice mechanisms to accelerate holistic ESG adoption among regulated entities. Ultimately, it formulates actionable, evidence-based recommendations for MCMC's internal policy, regulatory enforcement, and external engagement, strengthening ESG integration, corporate accountability, and sustainable development within the C&M industry.

RESEARCH OBJECTIVES

Researchers are invited to submit proposals guided by the following overarching Research Objectives (ROs) and Research Questions (RQs):

RO 1: To evaluate the current state of Environmental, Social, and Governance (ESG) integration within Malaysia’s C&M industry by examining the effectiveness of MCMC’s existing sustainability-related policies and identifying key drivers and barriers to ESG adoption across the industry.

- RQ 1.1: What is the current level and scope of ESG adoption among major players in the C&M industry, and how does it vary across different sub-sectors?

- RQ 1.2: In what ways have MCMC's existing policies and initiatives influenced ESG practices, performance, and reporting within the industry?

- RQ 1.3: What are the key challenges and enabling factors identified by stakeholders—including regulators, industry leaders, and civil society—in advancing ESG integration?

RO 2: To identify and analyse global regulatory frameworks and support strategies that have successfully facilitated ESG integration in communications sectors, and assess their relevance and adaptability to the Malaysian context.

- RQ 2.1: What ESG-related regulatory instruments and facilitation strategies have been effectively implemented by communications regulators in other countries?

- RQ 2.2: How do these international approaches compare with MCMC's current strategies in terms of scope, enforcement mechanisms, and stakeholder engagement?

- RQ 2.3: What lessons and adaptable elements from global best practices can inform the design of a Malaysian ESG facilitation framework tailored to the C&M industry?

RO 3: To develop a comprehensive and actionable ESG facilitation framework for MCMC, incorporating internal policy reforms, regulatory enforcement tools, and collaborative stakeholder engagement strategies to drive sector-wide ESG impact.

- RQ 3.1: What internal policy reforms and capacity-building initiatives are needed within MCMC to effectively lead ESG integration efforts?

- RQ 3.2: What regulatory enforcement tools and incentive structures can be introduced to encourage ESG compliance among C&M industry players?

- RQ 3.3: How can MCMC foster sustained and collaborative engagement with industry stakeholders, government bodies, and civil society to co-create meaningful ESG outcomes across the sector?

(The researcher is invited to propose refinements to Research Objectives and Research Questions. However, the researcher is requested to provide an explanation of the anticipated value of the proposed changes and potential benefits of proposed refinements.)

SESG-2 – Transforming the Malaysian Communications and Multimedia Designated Industry Fora: A Governance and Operational Review

A. Research Field:

- Malaysian Communications and Multimedia Commission Environmental, Social, and Governance (ESG) adoption.

B. Research Focus:

- MCMC policy development and facilitation of designated industry fora sustainable business and governance practices adoption.

C. Research Sponsor:

- Forum Strategy Planning Department, Forum Governance and Coordination Division, MCMC.

D. Contact Person for Queries:

- Ms. Elvira Shamsuddin, Director II, Forum Strategy Planning Department, Forum Governance and Coordination Division, elvira@mcmc.gov.my

E. Research Category:

- Sustainability & Environment, Social, and Governance (SESG).
  This research aims to generate evidence that will aid in the development of MCMC policies and facilitation of SESG adoption within the C&M designated industry fora.

F. Target Groups

1. **Stakeholders Benefiting from the Research**
   The findings and recommendations derived from this research are anticipated to yield benefits for these stakeholder groups:

   i. **Malaysian Communications and Multimedia Commission (MCMC)**: To support evidence-based policy-making and enhance oversight of industry fora.

   ii. **Designated Industry Fora**: To enhance their effectiveness in managing their roles and functions.

- Communications and Multimedia Content Forum of Malaysia (Content Forum);
- Communications and Multimedia Consumer Forum of Malaysia (Consumer Forum)
- Malaysian Technical Standards Forum Bhd (MTSFB) and
- Postal Forum

iii. **Industry Players and Service Providers**: To foster active participation in self-regulation mechanisms aligned with industry needs.

iv. **General Public**: As end users, they stand to gain from more transparent, responsive, and accessible fora**.**

2. **Groups/Communities as Interviewees or Survey Respondents for the Research**
   To ensure a comprehensive and well-rounded perspective, the research will engage with a diverse range of stakeholders, who will constitute the primary interviewees or survey respondents:

i. **Representatives** from **Designated Industry Fora**: To provide internal perspectives on operations, engagement strategies, and challenges.

(Where appropriate, researchers are encouraged to propose refinements to respondent categories to better align with the research aims.)

RESEARCH PROBLEM/CONTEXT

Malaysia's Communications and Multimedia (C&M) industry has adopted self-regulation as a foundational principle under the Communications and Multimedia Act (CMA) 1998, with the Malaysian Communications and Multimedia Commission (MCMC) empowered to designate industry fora to support collaborative governance. These fora are tasked with developing voluntary codes of practice, representing stakeholder interests, and contributing to policy development in areas such as consumer protection, content standards, technical specifications, and access.

Among the designated fora, the Consumer Forum of Malaysia (CFM) and the Content Forum of Malaysia (CMCF) are public-facing entities registered under the Registrar of Societies (ROS). Their governance is shaped by the Societies Act 1966, which emphasises democratic participation, transparency, and inclusivity. These fora are expected to engage a broad spectrum of stakeholders, including consumers, industry players, and civil society, while maintaining independence from commercial or political interests.

In contrast, the Malaysian Technical Standards Forum Berhad (MTSFB) is incorporated as a Berhad (Bhd) under the Companies Act 2016, which imposes a corporate governance framework focused on fiduciary responsibility, board oversight, and financial accountability. While MTSFB is also designated by MCMC to develop technical codes and standards, its structure reflects a more formalised, business-oriented model that may offer operational efficiency but less public engagement.

The Postal Forum, designated under the Postal Services Act 2012, adds a new dimension to this landscape. Although not incorporated as a company, it shares functional similarities with CFM and CMCF—particularly in its public-facing role, open membership, and consumer-centric mandate.

Despite their formal mandates and structural diversity, there is limited understanding of how Malaysia's self-regulatory industry fora fulfil their dual roles:

1. As entities governed by either the Societies Act or the Companies Act, with distinct governance, accountability, and transparency requirements.
2. As policy-influencing bodies under the CMA and related legislation, responsible for shaping industry standards and protecting consumer interests.

The evolving regulatory environment, coupled with global shifts in governance norms, calls for a reassessment of these fora' relevance, effectiveness, and structural robustness. This study investigates the governance models, operational effectiveness, and public-facing roles of three key fora—CFM, CMCF, and the Postal Forum—and compares them with corporate-style models of other fora and organisations, both locally and internationally. While MTSFB may serve as a reference point, researchers are encouraged to extend their enquiry to a wider range of organisations. The study aims to identify how legal form, funding mechanisms, and stakeholder engagement practices influence the fora' ability to deliver on their mandates, and to contribute to ongoing policy discourse on strengthening self-regulation in Malaysia's communications and multimedia sector.

Comparative Table: Governance Models of Industry Fora in Malaysia

Forum	Legal Basis	Registration Type	Governance Model	Regulatory Expectations
Consumer Forum (CFM)	CMA 1998	Registrar of Societies (ROS)	Participatory, stakeholder-driven	Voluntary codes, consumer protection, public engagement
Content Forum (CMCF)	CMA 1998	Registrar of Societies (ROS)	Participatory, stakeholder-driven	Content regulation, self-regulation via Content Code
Postal Forum	Postal Services Act 2012	Designated statutory body	Public-facing, advisory	Universal service planning, compliance reporting, service quality assurance
Malaysian Technical Standards Forum Bhd (MTSFB)	CMA 1998 + Companies Act 2016	Public Limited Company (Berhad)	Corporate, board-governed	Technical standards development, formal reporting, collaboration with SIRIM/Standards Malaysia

RESEARCH AIMS

This study aims to critically evaluate the governance structures, operational effectiveness, and public-facing roles of Malaysia's self-regulatory industry fora within the C&M industry. By examining fora registered under the Registrar of Societies (ROS)—namely the Consumer Forum of Malaysia (CFM), the Content Forum of Malaysia (CMCF), and the Postal Forum—the research compares these with the corporate-style governance models of other local and international fora and organisations. It seeks to understand how legal form, regulatory expectations, and stakeholder engagement mechanisms influence each forum's ability to fulfil its dual mandate:

1. As a governed entity under either the Societies Act 1966, the Postal Services Act 2012, or the Companies Act 2016.
2. As a policy-influencing body under the CMA 1998.

The research further aims to contribute to policy discourse on enhancing the accountability, transparency, and effectiveness of self-regulatory frameworks in Malaysia, particularly in light of evolving governance norms and sectoral challenges.

RESEARCH OBJECTIVES

Researchers are invited to submit proposals guided by the following overarching Research Objectives (ROs) and Research Questions (RQs):

RO 1: Evaluate the governance structures, legal forms, and regulatory compliance of Malaysia's self-regulatory industry fora.

- RQ 1.1: How do the governance structures and legal forms (ROS vs. Companies Act) influence the effectiveness and accountability of Malaysia's self-regulatory fora?

- RQ 1.2: To what extent do existing fora comply with their regulatory obligations under their respective legal frameworks?

RO 2: Assess the alignment of forum activities with the objectives of the CMA and Postal Services Act, and identify best practices from international models.

- RQ 2.1: To what extent do the activities and governance processes of Malaysian fora align with the objectives of the CMA and the Postal Services Act?

- RQ 2.2: What lessons can be drawn from international models of self-regulation to improve the Malaysian context?

RO 3: Examine alternative governance structures, including MTSFB, and propose a transformation framework to enhance forum effectiveness.

- RQ 3.1: How does the corporate-style governance of MTSFB compare with the society-based models in terms of operational efficiency and stakeholder engagement?

- RQ 3.2: What structural or procedural reforms are needed to strengthen the role of self-regulatory fora in Malaysia's C&M industry?

(The researcher is invited to propose refinements to Research Objectives and Research Questions. However, the researcher is requested to provide an explanation of the anticipated value of the proposed changes and potential benefits of proposed refinements.)

SESG-3 – Enhancing the Financial Sustainability of Designated Industry Fora in Malaysia: Exploring Alternative Mechanisms for Long-Term Viability

A. Research Field:

- Communications and Multimedia Ecosystem Sustainability & Environment, Social, and Governance (SESG) adoption.

B. Research Focus:

- Policy development and facilitation of designated industry fora sustainable business and governance practices adoption.

C. Research Sponsor:

- Forum Strategy Planning Department, Forum Governance and Coordination Division, MCMC.

D. Contact Person for Queries:

- Ms. Elvira Shamsuddin, Director II, Forum Strategy Planning Department, Forum Governance and Coordination Division, elvira@mcmc.gov.my

E. Research Category:

- Sustainability & Environment, Social, and Governance (SESG).
  This research aims to generate evidence that will aid in the development of relevant policies supporting and facilitating corporate governance and stakeholder accountability within the designated industry fora.

F. Target Groups

1. **Stakeholders Benefiting from the Research**
   The findings and recommendations derived from this research are anticipated to yield benefits for these stakeholder groups:

   i. **Malaysian Communications and Multimedia Commission (MCMC)**: To support evidence-based policy-making and enhance oversight of industry fora.

   ii. **Designated Industry Fora**: To enhance their effectiveness in managing their roles and functions.

- Communications and Multimedia Content Forum of Malaysia (Content Forum);

- Communications and Multimedia Consumer Forum of Malaysia (Consumer Forum)
- Malaysian Technical Standards Forum Bhd (MTSFB) and
- Postal Forum

iii. **Industry Players and Service Providers**: To foster active participation in self-regulation mechanisms aligned with industry needs.

iv. **General Public**: As end users, they stand to gain from more transparent, responsive, and accessible fora**.**

2. Groups/Communities as Interviewees or Survey Respondents for the Research

To ensure a comprehensive and well-rounded perspective, the research will engage with a diverse range of stakeholders, who will constitute the primary interviewees or survey respondents:

1. **Representatives from Designated Industry Fora**: To provide internal perspectives on operations, engagement strategies, and challenges.

(Where appropriate, researchers are encouraged to propose refinements to respondent categories to better align with the research aims.)

RESEARCH PROBLEM/CONTEXT

Under the Communications and Multimedia Act (CMA) and the Postal Services Act (PSA), the Malaysian Communications and Multimedia Commission (MCMC) recognises four designated industry fora: the Consumer Forum[21], Content Forum[22], Technical Standards Forum[23], and Postal Forum[24]. These fora are central to promoting industry self-regulation, contributing to the development of codes, standards, and guidelines that align with national policy objectives.

Despite their strategic importance, the long-term viability of these fora is increasingly challenged by their **heavy reliance on annual regulatory grants**. While they generate limited revenue through membership fees, this income is insufficient to sustain operations or support long-term initiatives. This overdependence raises concerns about their **financial resilience**, **institutional**

[21] Communications and Multimedia Consumer Forum of Malaysia (Consumer Forum). *About Us.* Accessed on 14 July 2025 at <https://cfm.my/?lang=en>

[22] Communications and Multimedia Content Forum of Malaysia (Content Forum). *About Us.* Accessed on 14 July 2025 at <https://contentforum.my/about-us/>

[23] Malaysian Technical Standards Forum Berhad (MTFSB) *About Us.* Accessed on 14 July 2025 at <https://mtsfb.org.my/about-us/>

[24] Postal Forum. *About Us.* Acessed on 14 July 2025 at <https://postalforum.my/about-us>

independence, and ability to function autonomously in a dynamic regulatory environment.

The absence of a **diversified and sustainable financial model** limits the fora's capacity to scale initiatives, innovate, and respond effectively to emerging sectoral challenges. Key indicators of financial sustainability—such as **revenue diversity**, **operating reserve ratio**, **dependency ratio**, and **liquidity**—highlight structural vulnerabilities that could undermine their effectiveness and credibility over time.

Given the critical role of these fora in advancing sectoral governance and public interest outcomes, there is an urgent need to explore **alternative funding strategies and institutional mechanisms**. This research aims to fill that gap by identifying and evaluating viable financial models—both domestic and international—that can enhance the long-term sustainability of Malaysia's industry fora, while preserving their regulatory integrity and alignment with national objectives.

RESEARCH AIMS

The primary aim of this research is to explore and evaluate alternative financial strategies that can strengthen the long-term sustainability of industry fora in Malaysia. Specifically, the study intends to support the Forum Governance and Coordination Division (FGCD) in fulfilling its mandate to ensure that these fora are financially secure and that their resources are managed effectively.

By examining viable funding models—both local and international—the research seeks to identify practical approaches that can reduce the fora's dependency on annual MCMC grants. The ultimate goal is to inform policy and operational decision-making, enabling the fora to achieve greater self-sufficiency, financial resilience, and effectiveness in carrying out their regulatory and sectoral roles.

RESEARCH OBJECTIVES

Researchers are invited to submit proposals guided by the following overarching Research Objectives (ROs) and Research Questions (RQs):

RO 1: To examine financial strategies and funding models employed by civil society organisations and self-regulatory bodies, both in Malaysia and internationally, for potential adaptation by local industry fora.

- RQ 1.1: Which alternative funding mechanisms are used by other civil society organisations and self-regulatory bodies which operate in a similar manner to Malaysia's four designated industry fora?

- RQ 1.2: What factors influence the adaptability of these models to Malaysia's designated industry fora?

RO 2: To identify and propose viable alternative funding mechanisms that can strengthen the financial sustainability of industry fora in Malaysia.

- RQ 2.1: What alternative funding sources (e.g. service fees, partnerships, grants) are available to the designated industry fora?

- RQ 2.2: What are the strengths and limitations of each funding mechanism in terms of sustainability and autonomy?

RO 3: To evaluate the current operational activities of the fora and identify areas with potential for monetisation or revenue generation, without compromising their regulatory integrity.

- RQ 3.1: What current activities or services offered by the fora have potential for monetisation?

- RQ 3.2: What are the risks of monetising these activities in terms of regulatory neutrality and public trust?

RO 4: To assess the feasibility, risks, and alignment of alternative funding options with national policy goals and stakeholder expectations.

- RQ 4.1: How do proposed funding models align with national policy goals and MCMC's strategic direction?

- RQ 4.2: What are the perceived risks and concerns among stakeholders regarding alternative funding models?

RO 5: To develop practical recommendations for MCMC and the fora to reduce dependence on government grants and enhance long-term financial resilience.

- RQ 5.1: What targeted and general policy recommendations can be implemented to reduce grant dependency while ensuring continued regulatory effectiveness and stakeholder trust?

(The researcher is invited to propose refinements to Research Objectives and Research Questions. However, the researcher is requested to provide an explanation of the anticipated value of the proposed changes and potential benefits of proposed refinements.)

SESG-4 – A Study on Public Utility Perception and Legal Safeguards for Fibre Infrastructure in Sabah: Addressing Vandalism, Theft, and Community Awareness

A. Research Field:

- Connectivity, Digital Inclusion and ICT4D (Information and Communication Technologies for Development).

B. Research Focus:

- Assessment of governance, legal frameworks, and community engagement strategies to enhance infrastructure resilience and public trust in digital utilities.

C. Research Sponsor:

- Celcom Timur (Sabah) Sdn. Bhd.

D. Contact Person for Queries:

- Mr. Loy Cheah Cheong, Head of Planning & Design, Network Planning & Design Network, Network Division, Celcom Timur (Sabah) Sdn. Bhd., loy@ctsabah.com.my

E. Research Category:

- Sustainability & Environment, Social, and Governance.
  This research contributes to the Sustainability and ESG agenda by examining governance mechanisms, community engagement, and legal safeguards that support the long-term resilience and public trust in Sabah's telecommunications infrastructure.

F. Target Groups

1. **Stakeholders Benefiting from the Research:**
   The findings and recommendations derived from this research are anticipated to yield benefits for these stakeholder groups:

   i. **Local Communities and Leaders** will benefit from improved service reliability and community empowerment through awareness and engagement in protecting infrastructure.

ii. **School Communities (Students and Teachers)** will have access to evidence-based insights to support public education and advocacy campaigns on infrastructure protection and digital responsibility.

iii. **Media and Public Advocacy Platforms** will have access to evidence-based insights to support public education and advocacy campaigns on infrastructure protection and digital responsibility.

iv. **Sabah Telecommunications Companies Network Operations and Legal Teams** will be equipped with data to support operational planning, risk mitigation, and legal reform, leading to reduced costs and improved service continuity.

v. **MCMC, Enforcement Agencies and Local Authorities** will benefit from clearer legal frameworks and enforcement guidelines, enhancing their capacity to respond effectively to infrastructure-related offences.

3. Groups/Communities as Interviewees or Survey Respondents for the Research

To ensure a comprehensive and well-rounded perspective, the research will engage with a diverse range of stakeholders, who will constitute the primary interviewees or survey respondents:

i. **Local Community Members and Leaders**
Can share lived experiences and perceptions, contributing to a better understanding of community-level challenges and attitudes.

ii. **Students and Teachers**
Can provide insights into how service disruptions affect learning and how they perceive the role of fibre infrastructure in education.

iii. **Enforcement Officers (MCMC, Police, Agencies and Local Authorities)**
Can reflect on enforcement challenges and gaps, helping to shape more effective legal and operational responses

iv. **Sabah Telecommunications Companies Network Operations and Legal Teams**
Can contribute operational and legal perspectives on infrastructure vulnerabilities, enforcement limitations, and potential improvements.

(Where appropriate, researchers are encouraged to propose refinements to respondent categories to better align with the research aims.)

RESEARCH PROBLEM/CONTEXT

In Sabah, fibre infrastructure—including poles, underground cables, and access boxes—faces persistent threats from vandalism, theft, and accidental damage. While some incidents are opportunistic or unintentional, a growing number are driven by economic motives, with high-value components such as copper and fibre cables targeted for resale. In June 2024, police arrested two individuals in Tawau for stealing communication cables, disrupting services to over 1,000 users and highlighting the operational and societal impact of such crimes (Bernama, 2024)[25].

This issue is not isolated to telecommunications. Sabah has also seen similar attacks on water and electricity infrastructure, often perpetrated by syndicates or individuals seeking profit from public assets (Lee, 2025)[26]. These cross-sector incidents point to a broader systemic vulnerability in how infrastructure is protected and perceived.

Compounding the problem is the lack of consistent enforcement and deterrence. While arrests do occur, existing penalties under laws such as the Communications and Multimedia Act 1998 or local by-laws may not reflect the full extent of the disruption caused. Fibre infrastructure is still widely viewed as private corporate property rather than a public utility, reducing community vigilance and weakening the social contract around its protection.

Environmental factors further exacerbate these vulnerabilities. Sabah's exposure to seasonal monsoons, flooding, and landslides can physically damage infrastructure, delay repairs, and leave assets exposed—creating windows of opportunity for theft or further degradation. These risks are especially pronounced in rural and remote areas where access and monitoring are limited.

There is an urgent need to reframe fibre infrastructure as a public utility deserving of legal protection and public respect. This study aims to investigate public perceptions, assess the impact of infrastructure damage, evaluate enforcement practices, and examine how environmental risks compound existing threats. It will also benchmark legal and community protection models from other sectors and jurisdictions, ultimately recommending targeted interventions to strengthen deterrence and safeguard Sabah's digital backbone.

[25] Bernama. (2024, June 3). *Two men nabbed for stealing communication cables in Tawau*. Malaysian National News Agency. Accessed on 29 July 2025 at <https://www.bernama.com/en/news.php?id=2267894>
[26] Lee, S. (2025, February 17). Catch the real culprits. The Star. Accessed on 29 July 2025 at <https://www.thestar.com.my/news/nation/2025/02/17/catch-the-real-culprits>

RESEARCH AIMS

To strengthen the protection and resilience of fibre infrastructure in Sabah by investigating public utility perception, evaluating legal and enforcement frameworks, and identifying strategies for community awareness and risk mitigation.

RESEARCH OBJECTIVES

Researchers are invited to submit proposals guided by the following overarching Research Objectives (ROs) and Research Questions (RQs):

RO 1: To examine public awareness, perception, and the socio-economic impact of fibre infrastructure in Sabah, particularly in the context of vandalism and theft.

- RQ 1.1: How do residents in different regions of Sabah perceive fibre infrastructure—private asset or public utility?
- RQ 1.2: What factors influence public understanding and attitudes toward fibre infrastructure?
- RQ 1.3: Are there notable differences in awareness between urban and rural communities?
- RQ 1.4: What are the direct and indirect costs of infrastructure damage due to vandalism and theft?
- RQ 1.5: How do service disruptions affect access to education, healthcare, and economic activities?
- RQ 1.6: Which areas or types of infrastructure are most frequently targeted?

RO 2: To evaluate the effectiveness of legal frameworks and benchmark best practices for protecting fibre infrastructure.

- RQ 2.1: What legal provisions currently exist to deter vandalism and theft of fibre infrastructure?
- RQ 2.2: How frequently are these laws enforced, and what are the outcomes of enforcement actions?
- RQ 2.3: Are current penalties proportionate to the impact of the offences?

- RQ 2.4: What community engagement and legal protection models are used in other Malaysian utility sectors?

- RQ 2.5: What international best practices exist for protecting high-value infrastructure?

- RQ 2.6: What lessons can be adapted for Sabah's telecommunications context?

RO 3: To propose targeted interventions to enhance the resilience and protection of fibre infrastructure.

- RQ 3.1: What types of public education campaigns are most effective in changing infrastructure-related behaviours?

- RQ 3.2: What policy or regulatory changes could improve deterrence and enforcement?

- RQ 3.3: How can community-based monitoring or reporting mechanisms be integrated into protection strategies?

(The researcher is invited to propose refinements to Research Objectives and Research Questions. However, the researcher is requested to provide an explanation of the anticipated value of the proposed changes and potential benefits of proposed refinements.)

SESG-5 – SW110 Waste Management in Sabah: Bridging Infrastructure Gaps and Evaluating Localised Treatment Feasibility

A. Research Field:

- Sustainability and Environment

B. Research Focus:

- Assessment of environmental impact of technology and e-waste management policy and enforcement.

C. Research Sponsor:

- Celcom Timur (Sabah) Sdn. Bhd.

D. Contact Person for Queries:

- Mr. Mohd Hairul bin Abd. Majid@Siedin, Head of Central Business Resources, Central Business Resources Department, Finance Division, Celcom Timur (Sabah) Sdn. Bhd., hairul@ctsabah.com.my

E. Research Category:

- Sustainability & Environment, Social, and Governance.
  This research contributes to the Sustainability and ESG agenda by critically examining governance mechanisms, policy frameworks, and legal provisions surrounding e-waste management in Sabah, highlighting gaps between regulation and real-world enforcement.

F. Target Groups

1. **Stakeholders Benefiting from the Research:**
   The findings and recommendations derived from this research are anticipated to yield benefits for these stakeholder groups:

   i. **Local Communities and Leaders** will benefit from improved service reliability and community empowerment through awareness and engagement in protecting infrastructure.

   ii. **School Communities (Students and Teachers)** will have access to evidence-based insights to support public education and advocacy campaigns on infrastructure protection and digital responsibility.

iii. **Media and Public Advocacy Platforms** will have access to evidence-based insights to support public education and advocacy campaigns on infrastructure protection and digital responsibility.

iv. **Sabah Telecommunications Companies Network Operations and Legal Teams** will be equipped with data to support operational planning, risk mitigation, and legal reform, leading to reduced costs and improved service continuity.

v. **MCMC, Enforcement Agencies and Local Authorities (Including Department of Environment and Environment Protection Department, Sabah)** will benefit from clearer legal frameworks and enforcement guidelines, enhancing their capacity to respond effectively to infrastructure-related offences.

2. Groups/Communities as Interviewees or Survey Respondents for the Research

To ensure a comprehensive and well-rounded perspective, the research will engage with a diverse range of stakeholders, who will constitute the primary interviewees or survey respondents:

i. **West Malaysian SW110 Facility Operators**
Can provide operational, regulatory, and management insights to support benchmarking and feasibility analysis for Sabah's context.

ii. **Local Community Members and Leaders**
Can share lived experiences and perceptions, contributing to a better understanding of community-level challenges and attitudes.

iii. **Students and Teachers**
Can provide insights into how service disruptions affect learning and how they perceive the role of fibre infrastructure in education.

iv. **Enforcement Officers (MCMC, Police, Agencies and Local Authorities)**
Can reflect on enforcement challenges and gaps, helping to shape more effective legal and operational responses

v. **Sabah Telecommunications Companies Network Operations and Legal Teams**
Can contribute operational and legal perspectives on infrastructure vulnerabilities, enforcement limitations, and potential improvements.

(Where appropriate, researchers are encouraged to propose refinements to respondent categories to better align with the research aims.)

RESEARCH PROBLEM/CONTEXT

The development of a dedicated facility for the treatment and disposal of SW110 scheduled waste in Sabah has been mooted at the state government level, following the decision to devolve environmental management powers from the federal Department of Environment (DOE) to the Environment Protection Department Sabah (EPD Sabah)[27]. As announced by the Sabah Cabinet in August 2024, the state will assume full responsibility for scheduled waste management starting January 2026. SW110 waste, which includes electrical and electronic assemblies containing hazardous components such as batteries, printed circuit boards, and mercury switches, must be handled at licensed facilities that meet stringent environmental and safety standards under the Environmental Quality (Scheduled Wastes) Regulations 2005.

As of July 2025, the Department of Environment Malaysia's official registry confirms that no such licensed SW110 facility currently exists in Sabah. In the absence of this infrastructure, waste generators are compelled to engage licensed contractors from Peninsular Malaysia for the collection, transportation, and treatment of SW110 waste. This practice significantly increases operational expenditure, introduces logistical complexity, and contributes to higher carbon emissions. It also poses challenges for regulatory compliance, transparency in waste tracking, and the broader sustainability agenda.

Continuing with the current model may limit Sabah's ability to build local capacity, reduce environmental risks, and meet national ESG targets. Conversely, establishing a localised SW110 treatment facility presents opportunities for cost optimisation, improved compliance, and regional infrastructure development. Even if a full-scale facility is deemed unfeasible, the research can illuminate alternative strategies such as decentralised waste hubs, mobile treatment units, or enhanced digital governance mechanisms.

This context underscores the strategic importance of evaluating the feasibility of a localised solution that aligns with regulatory standards and supports Sabah's transition towards a more sustainable and equitable waste management ecosystem

[27] Environment Protection Department Sabah. (2024, August 28). *Sabah to manage its own scheduled waste by 2026*. Accessed on 30 July 2025 at <https://epd.sabah.gov.my/v3/sabah-to-manage-its-own-scheduled-waste-by-2026/>

RESEARCH AIMS

This research aims to evaluate the feasibility and strategic importance of establishing a localised scheduled waste treatment facility in Sabah, with a specific focus on SW110 waste as defined under the Environmental Quality (Scheduled Wastes) Regulations 2005[28]. The study will examine the technical, environmental, regulatory, and financial considerations required to support the development of a facility that complies with Department of Environment (DOE) standards.

The outcome of this research will provide a comprehensive, evidence-based recommendation for relevant authorities, industry stakeholders, and policymakers. It will also explore alternative models and improvements in the event that a full-scale SW110 facility is not feasible, thereby contributing to informed decision-making, enhanced compliance, and long-term sustainability in scheduled waste management across Sabah.

RESEARCH OBJECTIVES

Researchers are invited to submit proposals guided by the following overarching Research Objectives (ROs) and Research Questions (RQs):

RO 1: To analyse the potential for cost reduction through localised treatment, including savings on transportation, third-party handling fees, and storage.

- RQ 1.1: What are the current cost components associated with SW110 waste handling in Sabah?
- RQ 1.2: How much cost saving could be achieved through localised or decentralised treatment options?
- RQ 1.3: What are the financial implications for waste generators and contractors under various treatment models (e.g. full facility, mobile units, regional hubs)?

RO 2: To identify gaps, limitations, and challenges in existing scheduled waste management practices within Sabah.

- RQ 2.1: What are the operational and regulatory limitations in Sabah's current scheduled waste management system?

[28] Department of Environment Malaysia. (2005). Environmental Quality (Scheduled Wastes) Regulations 2005 – P.U. (A) 294/2005. Accessed on 30 July 2025 at <https://ewaste.doe.gov.my/wp-content/uploads/2020/12/Environmental_Quality_Scheduled_Wastes_Regulations_2005_-_P.U.A_294-2005.pdf>

- RQ 2.2: What barriers exist in establishing a local treatment facility (e.g. technical, legal, environmental)?

- RQ 2.3: What short- or medium-term improvements could be implemented to strengthen existing practices while a full facility is being evaluated?

- RQ 2.4: How do stakeholders perceive the effectiveness and adaptability of the current system?

RO 3: To benchmark Sabah against West Malaysian states with successful SW110 treatment facilities.

- RQ 3.1: What are the key success factors in West Malaysian states with SW110 facilities?

- RQ 3.2: How do Sabah's conditions compare in terms of waste volume, infrastructure, and policy support?

- RQ 3.3: What lessons can be adapted for Sabah's context?

RO 4: To recommend an implementation framework and policy considerations for a sustainable, DOE-compliant facility in Sabah.

- RQ 4.1: What are the critical components of a DOE-compliant SW110 treatment facility?

- RQ 4.2: What policy instruments, incentives, or partnerships could support short- to medium-term solutions?

- RQ 4.3: How can stakeholder collaboration be structured to support both interim improvements and long-term infrastructure development?

- RQ 4.4: What decision-making criteria should be used to assess the feasibility of different treatment models?

(The researcher is invited to propose refinements to Research Objectives and Research Questions. However, the researcher is requested to provide an explanation of the anticipated value of the proposed changes and potential benefits of proposed refinements.)

SESG-6 – Evaluating the Relevance of Regulatory Definitions in the General Consumer Code (GCC) Amidst Malaysia's Evolving Communications and Multimedia Landscape

A. Research Field:

- Sustainable Business and Governance Practices.

B. Research Focus:

- Corporate governance and stakeholder accountability.

C. Research Sponsor:

- The Communications and Multimedia Consumer Forum of Malaysia (CFM).

D. Contact Person for Queries:

- Ms. Nur Aishah Mohamed Yussof, Executive, Data Analyst, aishah@cfm.my
- Ms. Nadia Fakarrudy, Executive, Policy, Development and Governance, nadia.fakarrudy@cfm.my
- Ms. Hayyie Mubarak Anwar Mubarak, Executive, Complaint and Compliance, hayyie@cfm.my

E. Research Category:

- Sustainability & Environment, Social, and Governance.
  This study aims to determine the relevance of the General Consumer Code of Practice for the Communications and Multimedia Industry Malaysia (GCC) developed by the CFM in ensuring sustainable business and consumer practices through sound corporate governance and stakeholder accountability.

F. Target Groups

1. Stakeholders Benefiting from the Research

The findings and recommendations derived from this research are anticipated to yield benefits for these stakeholder groups:

i. **Consumers**

- Urban and rural users of mobile, broadband, and multimedia services.
- Individuals who have lodged complaints or experienced service issues.
- Vulnerable groups (e.g., elderly, disabled, low-income users) to assess inclusivity of definitions.

ii. **Service Providers**
- Licensees under the Malaysian Communications and Multimedia Commission (MCMC).
- Benefit from clear guidelines on consumer engagement, complaint resolution, and service standards.
- Helps maintain industry credibility and regulatory compliance.

iii. **Regulators and Policy-makers**
- MCMC and the Communications and Multimedia Consumer Forum of Malaysia (CFM).
- Use the GCC as a benchmark for monitoring industry practices and enforcing consumer protection standards.

iv. **Industry Stakeholders**
- Includes content providers, network operators, and digital service aggregators.
- Benefit from a harmonised framework that supports self-regulation, corporate governance, and stakeholder accountability.

3. Groups/Communities as Interviewees or Survey Respondents for the Research

To ensure a comprehensive and well-rounded perspective, the research will engage with a diverse range of stakeholders, who will constitute the primary interviewees or survey respondents:

i. **Consumers**
- Urban and rural users of mobile, broadband, and multimedia services.
- Individuals who have lodged complaints or experienced service issues.
- Vulnerable groups (e.g., elderly, disabled, low-income users) to assess inclusivity of definitions.

ii. **Service Providers**
- Compliance officers or regulatory affairs managers from telcos and ISPs.
- Customer service heads or complaint resolution teams.
- Legal or policy advisors involved in interpreting the GCC.

iii. **Regulators and Policy Makers**
- Representatives from MCMC and CFM.
- Officers involved in code development, enforcement, or review.

iv. **Industry Associations and Advocacy Groups**

- Consumer rights organisations.
- Industry bodies representing telecommunications and multimedia service providers.
- Legal experts or academics specialising in communications law and policy.

(Where appropriate, researchers are encouraged to propose refinements to respondent categories to better align with the research aims.)

RESEARCH PROBLEM/CONTEXT

The General Consumer Code of Practice for the Communications and Multimedia Industry Malaysia (GCC)[29] serves as a foundational regulatory instrument for safeguarding consumer rights and guiding service providers. Its effectiveness depends heavily on the clarity and relevance of its definitions, which shape expectations, obligations, and enforcement mechanisms across the industry.

Malaysia's telecommunications and multimedia landscape has evolved rapidly, driven by emerging technologies such as Voice over IP (VoIP), Internet of Things (IoT), and bundled digital services. These innovations have blurred traditional service categories and introduced hybrid offerings, creating consumer relationships that were not fully anticipated when the GCC was first drafted.

Developed by the designated industry forum under Section 95 of the Communications and Multimedia Act 1998 (CMA), the GCC embodies a collaborative regulatory framework. In view of evolving service models and shifting consumer dynamics, we propose targeted amendments to the GCC's regulatory definitions pursuant to Section 102 of the CMA. These revisions are intended to ensure that the Code remains responsive, enforceable, and aligned with current industry realities.

RESEARCH AIMS

This study aims to evaluate the relevance and adequacy of existing definitions within the General Consumer Code (GCC), developed by the CFM, in the context of Malaysia's evolving telecommunications and multimedia ecosystem. It will assess how outdated or ambiguous definitions may impact industry compliance, regulatory clarity, consumer understanding, and ultimately, the sustainability of business and consumer practices.

[29] Communications and Multimedia Consumer Forum of Malaysia. (2003). *General Consumer Code of Practice for the Communications and Multimedia Industry Malaysia*. Accessed on 4 August 2025 at <https://www.mcmc.gov.my/skmmgovmy/files/attachments/GeneralConsumerCode.pdf>

By analysing cases or patterns where unclear definitions have led to inconsistent interpretations or service complaints, the research will identify definitional gaps affecting both service providers and end users. The expected value of this research lies in its potential to inform a more precise and future-proof revision of the GCC, particularly in the Definitions and Interpretation section and other relevant parts of the Code.

In doing so, the study supports the organisation's strategic goals of enhancing corporate governance, stakeholder accountability, and consumer protection. It contributes to key performance indicators such as improved industry compliance, faster complaint resolution, and greater stakeholder satisfaction with the Code. Ultimately, the research reinforces the organisation's role in facilitating industry self-regulation while promoting responsible and sustainable business practices.

RESEARCH OBJECTIVES

Researchers are invited to submit proposals guided by the following overarching Research Objectives (ROs) and Research Questions (RQs):

RO 1: To assess the relevance and adequacy of existing definitions in the GCC in the context of emerging technologies and service models.

- RQ 1.1: To what extent are the current definitions in the GCC aligned with technological and service innovations such as VoIP, IoT, and bundled digital services?
- RQ 1.2: How do these definitions accommodate hybrid service offerings and evolving consumer relationships?

RO 2: To examine how definitional ambiguity or outdated terminology affects industry compliance, regulatory enforcement, and consumer understanding.

- RQ 2.1: How do ambiguous or outdated definitions in the GCC affect regulatory compliance among service providers?
- RQ 2.2: What challenges do regulators face in enforcing the GCC due to definitional inconsistencies?
- RQ 2.3: In what ways do unclear definitions impact consumer understanding and contribute to service-related complaints or disputes?

RO 3: To identify specific cases or patterns where unclear definitions have led to inconsistent interpretations or service complaints.

- RQ 3.1: What definitional gaps or inconsistencies can be identified through case analysis or stakeholder feedback?

- RQ 3.2: How frequently do these gaps result in disputes or non-compliance, and what are the typical outcomes?

RO 4: To propose recommendations for refining the GCC to enhance regulatory clarity, corporate governance, and stakeholder accountability.

- RQ 4.1: What definitional revisions could improve clarity and reduce interpretational discrepancies?

- RQ 4.2: How can these revisions support stronger corporate governance and stakeholder accountability?

RO 5: To evaluate how improved definitions could contribute to sustainable business practices and consumer protection.

- RQ 5.1: What are the potential impacts of clearer and more current definitions on industry self-regulation?

- RQ 5.2: How might these improvements enhance consumer protection and trust in service delivery?

(The researcher is invited to propose refinements to Research Objectives and Research Questions. However, the researcher is requested to provide an explanation of the anticipated value of the proposed changes and potential benefits of proposed refinements.)

SESG-7 – Strengthening Business Stability in Malaysian Broadcasting through Sustainable Local Content Creation: A Strategic Framework for Producers and Regional Broadcasters

A. Research Field:

- Sustainable Business and Governance Practices.

B. Research Focus:

- Sustainable local content creation.

C. Research Sponsor:

- Sarawak Media Group Sdn. Bhd.

D. Contact Person for Queries:

- Ms. Madiana Binti Roland Nawe Salas, Acting Head, Business Development, Sarawak Media Group Sdn. Bhd, madiana@smg.my

E. Research Category:

- Sustainability & Environment, Social, and Governance.
  This study aims to understand the factors contributing to and impeding the sustainability of local Malaysian content creation towards ensuring the sustainability of Malaysia broadcasters and local content creators.

F. Target Groups

1. **Stakeholders Benefiting from the Research**
   The findings and recommendations derived from this research are anticipated to yield benefits for these stakeholder groups:

   i. **Local Content Creators**
   Independent filmmakers, production houses, writers, and creative professionals who will gain insights into sustainable practices, funding opportunities, and market strategies.

   ii. **Broadcasting Companies**
   TV networks and radio stations that rely on local content to build brand identity, retain audiences, and ensure long-term business viability.

   iii. **Digital Content Platforms**
   OTT services and online platforms that distribute local content.

iv. **Media Investors**
Private equity firms, venture capitalists, and public funding bodies seeking viable investment opportunities in the creative sector.

v. **Government Agencies**
Entities like FINAS, MCMC, and the Ministry of Communications that can use the findings to inform policy, funding schemes, and regulatory frameworks.

2. Groups/Communities as Interviewees or Survey Respondents for the Research

To ensure a comprehensive and well-rounded perspective, the research will engage with a diverse range of stakeholders, who will constitute the primary interviewees or survey respondents:

i. **Local Content Producers**
Creators involved in various formats (docu-series, drama, short films, etc.) who can share firsthand experiences of operational and market challenges.

ii. **Broadcast Executives**
Decision-makers in media companies who can provide insights into content acquisition, audience strategies, and business models.

iii. **Digital Content Platforms Executives.**
Decision-makers serving in OTT services and online platforms that distribute local content.

iv. **Funding Bodies / Investors**
Representatives from organisations that fund or invest in media projects, offering perspectives on risk, ROI, and funding criteria..

v. **Policy Makers / Regulators**
Officials from government agencies who can discuss existing support mechanisms and policy gaps.

vi. **Audience Representatives**
Focus groups or survey participants who can help define what makes local content engaging, relevant, and impactful.

(Where appropriate, researchers are encouraged to propose refinements to respondent categories to better align with the research aims.)

RESEARCH PROBLEM/CONTEXT

Malaysia's broadcasting landscape is undergoing rapid transformation, driven by digital migration and intensified competition from global streaming platforms. While local content remains vital for cultural representation and audience engagement, its creators face persistent challenges in monetisation, visibility, and scalable distribution. Regional broadcasters—many of which operate under public or state ownership—play a crucial role in amplifying local voices and sustaining creative ecosystems. However, they too struggle to balance public service mandates with commercial viability.

Despite institutional support, there is a lack of integrated strategies that align content creation with sustainable business models. The absence of clear benchmarks for success and limited access to diversified revenue streams further hinder the growth of both producers and broadcasters. This research seeks to address these systemic challenges by developing a strategic framework that supports the creation, promotion, and monetisation of high-quality local content, thereby strengthening the business stability of Malaysian broadcasters and empowering local producers.

RESEARCH AIMS

To develop a strategic framework that enhances the sustainability, visibility, and commercial viability of local content creation by strengthening the ecosystem of regional broadcasters and content producers in Malaysia.

RESEARCH OBJECTIVES

Researchers are invited to submit proposals guided by the following overarching Research Objectives (ROs) and Research Questions (RQs):

RO 1: To examine the operational and market challenges faced by regional broadcasters and local content creators.

- RQ 1.1: What are the key operational constraints faced by broadcasters in sourcing and airing local content?
- RQ 1.2: What market factors limit the visibility and monetisation of local content in regional contexts?

RO 2: To assess the effectiveness of existing support mechanisms (governmental, institutional, financial) for local content creation.

- RQ 2.1: What types of support do broadcasters and creators currently receive from agencies such as FINAS or state governments?

- RQ 2.2 – How well do these mechanisms align with the commercial needs of regional broadcasters and producers?

RO 3: To identify characteristics of successful local content in regional broadcasting contexts.

- RQ 3.1: What defines high-impact content for regional audiences (e.g. cultural relevance, language, storytelling)?

- RQ 3.2: How do audience engagement metrics differ between regional and national platforms?

RO 4: To propose a sustainable content creation and distribution model for regional broadcasters and local producers.

- RQ 4.1: What business models can support both public service and commercial objectives?

- RQ 4.2: How can digital platforms be leveraged to extend the reach and revenue of local content?

RO 5: To explore the role of regional broadcasters in nurturing local creative ecosystems.

- RQ 5.1: How do regional broadcasters contribute to talent development and industry growth?

- RQ 5.2: What partnerships or collaborations can enhance the scalability of local content?

(The researcher is invited to propose refinements to Research Objectives and Research Questions. However, the researcher is requested to provide an explanation of the anticipated value of the proposed changes and potential benefits of proposed refinements.)

SESG-8 – Closing the Data Gaps: Developing Locally Representative Emission Factors for Malaysia's Transportation Sector to Strengthen Climate Policy and Reporting

A. Research Field:

- Sustainable logistics and fleet decarbonisation in the postal and courier sector.

B. Research Focus:

- Assessment of pollutant emission factors associated with delivery and fleet operations in the postal and courier sector.

C. Research Sponsor:

- Pos Malaysia Berhad.

D. Contact Persons for Queries:

- Ms. Jean Chook, Senior Manager, Sustainability Reporting and ESG Strategy, Marketing, Communications and Sustainability Department, Pos Malaysia Berhad, yijean@pos.com.my

E. Research Category:

- Sustainability & Environment, Social, and Governance (SESG).
  This study contributes to the SESG agenda by examining environmental emission factors and measurement methodologies relevant to industries heavily dependent on transportation (e.g., logistics, delivery services), with a focus on the postal and courier sector's transition toward low-carbon fleet operations.

F. Target Groups

3. Stakeholders Benefiting from Research

The findings and recommendations derived from this research are anticipated to yield benefits for these stakeholder groups:

i. **Government agencies** involved in climate policy, transport regulation, and emissions reporting (e.g., Ministry of Transport, Department of Environment, MyHijau).

ii. **Policy-makers** responsible for designing fuel economy standards, carbon pricing mechanisms, and urban planning strategies.

iii. **Postal and courier service providers** seeking to transition to low-emission fleets and improve climate disclosures.

iv. **Corporate sustainability teams and ESG reporting units** aiming to align with IFRS S2 and other international frameworks.

v. **Urban planners and local authorities** interested in data-driven approaches to reduce transport-related pollution.

vi. **Environmental NGOs and advocacy groups** working on air quality, climate justice, and sustainable mobility.

4. Groups/Communities as Interviewees or Survey Respondents for the Research

To ensure a comprehensive and well-rounded perspective, the research will engage with a diverse range of stakeholders, who will constitute the primary interviewees or survey respondents:

i. **Fleet managers and logistics coordinators** from postal, courier, and transport companies (e.g., Pos Malaysia, DHL, J&T Express).

ii. **Technical officers and analysts** from government departments (e.g. Bursa Malaysia) responsible for emissions inventories and transport data.

iii. **Academics and researchers** specialising in transport emissions, environmental modelling, and climate policy.

iv. **Industry associations** such as the Malaysian Institute of Road Safety Research (MIROS), Federation of Malaysian Manufacturers (FMM), and logistics chambers.

(Where appropriate, researchers are encouraged to propose refinements to respondent categories to better align with the research aims.)

RESEARCH PROBLEM/CONTEXT

The transportation sector is the second-largest contributor to Malaysia's greenhouse gas (GHG) and air pollutant emissions. However, the country lacks accurate, locally representative emission factors. Existing data often rely on outdated or foreign-based factors that do not adequately reflect Malaysia's diverse vehicle fleet, fuel quality, driving behaviour, and climatic conditions. This data gap hampers the development of targeted climate policies, limits the accuracy of

national emissions inventories, and constrains alignment with international frameworks such as IFRS S2, IPCC Guidelines, and UNFCCC reporting requirements.

This research holds particular relevance for the postal, courier, freight and logistics sector, which is undergoing a significant transformation towards low-carbon operations. Organisations in this sector operate extensive fleets that contribute substantially to urban emissions, particularly through first- and last-mile delivery. Accurate, localised emission factors are essential for these companies to quantify their environmental impact, optimise fleet transition strategies, and meet evolving climate disclosure standards.

A more granular understanding of pollutant emission factors will enable courier companies to support transparent climate disclosures, improve operational efficiency, and align with national and international climate goals.

Beyond corporate applications, the research offers broader societal benefits to those outside of the transportation sector to quantify their Scope 3 emissions generated within the transportation sector. These include improved understanding of urban air quality, more informed transport and infrastructure planning, and reduced emissions through evidence-based policymaking—ultimately contributing to healthier, more sustainable cities.

RESEARCH AIMS

To identify and analyse data gaps in pollutant emission factors within Malaysia's transportation sector, benchmark current practices, and propose a locally representative framework that supports effective climate policy, emissions reporting, and international alignment.

RESEARCH OBJECTIVES

Researchers are invited to submit proposals guided by the following overarching Research Objectives (ROs) and Research Questions (RQs):

RO 1: To evaluate the current landscape of emission factors used for estimating transportation-related pollutants in Malaysia, and assess their alignment with international standards.

- RQ 1.1: What are the primary sources and methodologies currently used to derive emission factors for Malaysia's transportation sector?
- RQ 1.2: To what extent do these emission factors align with international standards such as IPCC Guidelines, EMEP/EEA protocols, and UNFCCC reporting requirements?

- RQ 1.3: What assumptions underpin the current emission factor models, and how do these affect their accuracy and relevance?

- RQ 1.4: How frequently are emission factors updated, how much it costs for updating these emission factors and what mechanisms exist for their revision or validation?

RO 2: To identify and analyse key data gaps and limitations in emission factor availability, consistency, and applicability across transport modes and pollutant types.

- RQ 2.1: What are the major gaps in emission factor data for different transport modes (e.g., road, rail, air, maritime) and different type of vehicles (e.g., motorcycle, MPV, panel van, 1-tonne truck, 5-tonne truck)?

- RQ 2.2: How do these gaps vary across pollutant types such as CO_2, NO_x, PM2.5, and VOCs?

- RQ 2.3: What challenges exist in collecting granular, mode-specific emission data in Malaysia?

- RQ 2.4: How do regional disparities (e.g., Peninsular vs East Malaysia) affect the availability and applicability of emission factors?

- RQ 2.5: How do different driving behavior affect the emission factors?

RO 3: To benchmark Malaysia's emission factor practices against regional and international best practices, particularly in comparable contexts.

- RQ 3.1: How do Malaysia's emission factor development and usage practices compare with those of countries with similar vehicle fleets, climate conditions, or development levels?

- RQ 3.2: What institutional or technical capacities enable other countries to maintain robust emission factor datasets?

- RQ 3.3: What lessons can be drawn from regional initiatives (e.g., ASEAN, APEC) in harmonising emission factor methodologies?

- RQ 3.4: What role do public-private partnerships play in emission factor development in benchmarked countries?

RO 4: To assess the implications of emission factor data gaps on climate policy, emissions reporting, and corporate disclosures.

- RQ 4.1: How do current data limitations impact national GHG inventories and climate policy instruments (e.g., fuel economy standards, carbon pricing)?

- RQ 4.2: What are the consequences for corporate climate disclosures, under IFRS S2?

- RQ 4.3: How do emission factor inaccuracies affect Malaysia's ability to meet its Nationally Determined Contributions (NDCs)?

- RQ 4.4: What are the risks of misreporting or underreporting emissions due to reliance on non-localised emission factors?

RO 5: To propose a framework for developing and adopting locally representative emission factors, including stakeholder engagement and integration into national systems.

- RQ 5.1: What are the essential components of a robust framework for emission factor development in Malaysia?

- RQ 5.2: How can stakeholders such as government agencies, academia, and industry be effectively involved in data collection and implementation?

- RQ 5.3: What technological and institutional enablers are needed to support continuous improvement of emission factor datasets?

- RQ 5.4: How can the proposed framework be integrated into existing national emissions reporting systems and climate governance structures?

(The researcher is invited to propose refinements to Research Objectives and Research Questions. However, the researcher is requested to provide an explanation of the anticipated value of the proposed changes and potential benefits of proposed refinements.)